Shale facies in the Mesoproterozoic Beetaloo Sub-basin, Northern Australia—Experiment-based assessment of depositional conditions for the world's oldest gas shale

Juergen Schieber, Sandra Menpes, and Tim J. Munson

ABSTRACT

Interpreting depositional settings of ancient shales relies on careful facies analysis, recognition of primary sedimentary structures and textures, and the "readout" they provide with regard to physical, chemical, and biological processes at the seabed. For Phanerozoic shales, modern analogues facilitate such analysis, but for Precambrian examples, nonactualistic scenarios need to be seriously considered, given that the biosphere was fundamentally dominated by prokaryotic microbes. Fortunately, in spite of their antiquity, many Precambrian shales show an abundance of sedimentary features that can be calibrated to process variables via creative use of experimental studies.

The Mesoproterozoic Velkerri Formation of northern Australia contains the shale-dominated carbonaceous Amungee Member, known for abundant hydrocarbon shows and producible shale gas. The dark-light "striped" appearance of these shales reflects alternating layers (millimeters to centimeters) of variable total organic carbon (TOC) content (>4% versus <2%, respectively). Some TOC-rich layers show a wavy-lenticular internal texture of anastomosing C-org and pyrite-rich streaks and laminae, textural attributes suggestive of benthic microbial mats. The majority of TOC-rich layers shows lenticular fabric, composed of sand-sized (0.1–1 mm) elongate fine-grained aggregates of variable composition. Aggregate size varies between layers;

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imbrication and inclined layering is suggestive of bedload transport. Flume studies of lenticular fabric suggest that these layers reflect seafloor erosion of soft muds in the form of sand-sized aggregates that were transported across the basin floor by bottom currents. In the context of the Amungee Member, the observed features imply that current events of varying strength eroded parts of the muddy seabed and distributed soft "clasts" across the basin via bottom current transport. Interspersed parallel laminae of coarse silt, and homogenous clay-silt layers with inclined fabric elements suggest bottom current transport of silt and flocculated muds, an interpretation that is consistent with flume investigations.

In shales, textural studies at the bed- and lamina scale provide insights into conditions of sediment transport and deposition. Through linkage to experimental work, parameters like current velocity and duration of transport events, can be estimated and support predictive models of shale accumulation even for "deep-time" strata.

INTRODUCTION

Fine-grained sedimentary rocks (shale, mudstone, etc.) are the dominant ingredient of the sedimentary record, providing the most complete stratigraphic archives for interpreting paleoclimate and paleo-oceanography (e.g., Reineck and Singh, 1980; Stanley, 1983; Allen, 1985; Bohacs, 1998; Warren et al., 1998; Bhattacharya and MacEachern, 2009; Macquaker et al., 2010b) and a subset of these rocks acts as source, reservoir, and seal for hydrocarbons. Technological advances in reservoirs engineering, such as hydrofracturing of horizontal wells, have opened up the vast energy resources contained in these rocks and have profoundly changed the worldwide outlook on fossil fuel production over the past three decades (Gold, 2014).

Interpreting the depositional setting of ancient shales relies on careful facies analysis, recognition of primary sedimentary structures and textures, and their interpretation in the context of the most likely physical, chemical, and biological processes at the seabed. When we consider sedimentary deposits of Mesozoic and even Paleozoic age, our facies evaluation is greatly helped by studies of modern analogues and our understanding of biological parameters. Uncertainty about analogue efficacy however, increases as we move further into "deep time." Once we enter the Precambrian, almost all of the inferences that could be made from fossil content and ichnology no longer apply, and we are increasingly forced to entertain nonactualistic scenarios. On the bright side, although shales appear to many as comparatively featureless and defying categorization in terms of depositional processes, careful examination invariably shows them to be rich in sedimentary features that can be calibrated (like their younger counterparts) to process variables via creative use of experimental studies. Precambrian shales are also essentially undisturbed by bioturbation, which generally results in the excellent preservation of primary depositional textures in unmetamorphosed and even low grade metamorphic successions (Schieber, 1986, 1989, 1990) and also maximizes the chances that organic matter (OM) at the sediment-water interface is preserved. In this paper we link detailed core descriptions and petrography of Mesoproterozoic shales from three drillholes in the Beetaloo Sub-basin of the McArthur Basin of the Northern Territory, Australia (Rawlings, 1999; Abbott et al., 2001; Cox et al., 2022; Crombez et al., 2023; Garrad, 2024; Soares et al., 2025), to flume analogue studies (Schieber et al., 2007a, 2010; Schieber, 2011, 2016a; Yawar and Schieber, 2017), in order to identify likely underlying physical, chemical, and biological processes and process parameters in this very old succession where the factors controlling OM preservation and source rock formation might be quite different from those in Phanerozic and younger counterparts.

Geologic Setting

The informally named greater McArthur Basin (Close, 2014) is a vast Palaeoproterozoic- to Mesoproterozoic-aged terrane extending over 180,000 km^2 from Western Australia, through the Northern Territory into northwestern Queensland (Figures 1, 2). It includes Palaeoproterozoic to Mesoproterozoic successions of the McArthur Basin sensu stricto, Birrindudu Basin, and the Tomkinson Province (Figure 1, inset), all of which are interpreted to be linked in the subsurface, based on geophysical and drilling evidence. The greater McArthur Basin comprises predominantly sedimentary successions that are unmetamorphosed and mostly flat-lying or gently folded. Deposition and subsequent inversion within the basin were strongly influenced by faults and complex fault zones with typically long histories (ca. 1800–1300 Ma) of reactivation and influence on basin architecture (Jarrett et al., 2022). The basin unconformably overlies Archaean and Paleoproterozoic rocks of the North Australian Craton and is unconformably overlain by Neoproterozoic–Palaeozoic cover successions (Ahmad et al., 2013).

The sedimentary successions of the McArthur Basin sensu stricto were subdivided by Rawlings (1999) into five basin-scale, nongenetic depositional "packages": in ascending order, the Palaeoproterozoic Redbank, Goyder, and Glyde packages; and the Mesoproterozoic Favenc and Wilton packages. The term "package" was originally defined as an amalgamation of lithostratigraphic units with similar ages, stratigraphic position, lithofacies or lithofacies associations, and style and composition of volcanism, and basin-fill geometry (Rawlings, 1999). These packages were subsequently informally extended by the Northern Territory Geological Survey (NTGS) across the greater McArthur Basin to include correlative successions within the Birrindudu Basin and Tomkinson Province (Close, 2014; Figures 1, 3). Except for the Goyder package, they are bounded by tectono-eustatic disconformities or unconformities that have regional extent across all, or much of the basin. In general, packages young toward the central depocentre (Beetaloo Sub-basin) and toward north–south-oriented major fault zones in the east, where the successions are inverted.

The Beetaloo Sub-basin (Figure 1) is a composite depocenter extending over an area of approximately 28,000 km^2 (Williams, 2019). It contains approximately 9000 m of Palaeoproterozoic to Mesoproterozoic sedimentary and minor volcanic rocks (Close, 2014) and is considered the principal depocenter for the Roper Group (Wilton package), which includes the economically significant Velkerri Formation of the upper Roper Group (Munson, 2016; Munson and Revie, 2018; Jarrett et al., 2022; references therein). The subbasin is bordered to the north by the northwest-trending Mallapunyah fault zone and is divided by the broadly north-trending Daly Waters Fault Zone (DWFZ) into two main depositional areas (Figures 1, 4). Other margins are more arbitrarily positioned on transitions to a condensed succession on surrounding platform areas; Williams (2019) placed them where the top of the Kyalla Formation at the top of the Roper Group was either missing or shallower than 400 m below the surface. Based on sedimentological studies (e.g., Munson, 2016) and isotopic studies of detrital zircons (e.g., Yang et al., 2018), sediment source areas for the Roper Group are interpreted to have been mainly from elevated crystalline basement areas to the present-day southeast and south. Garrad (2023, 2024) and Soares et al. (2025) interpreted the Roper Group in the subbasin as having been deposited during a period of

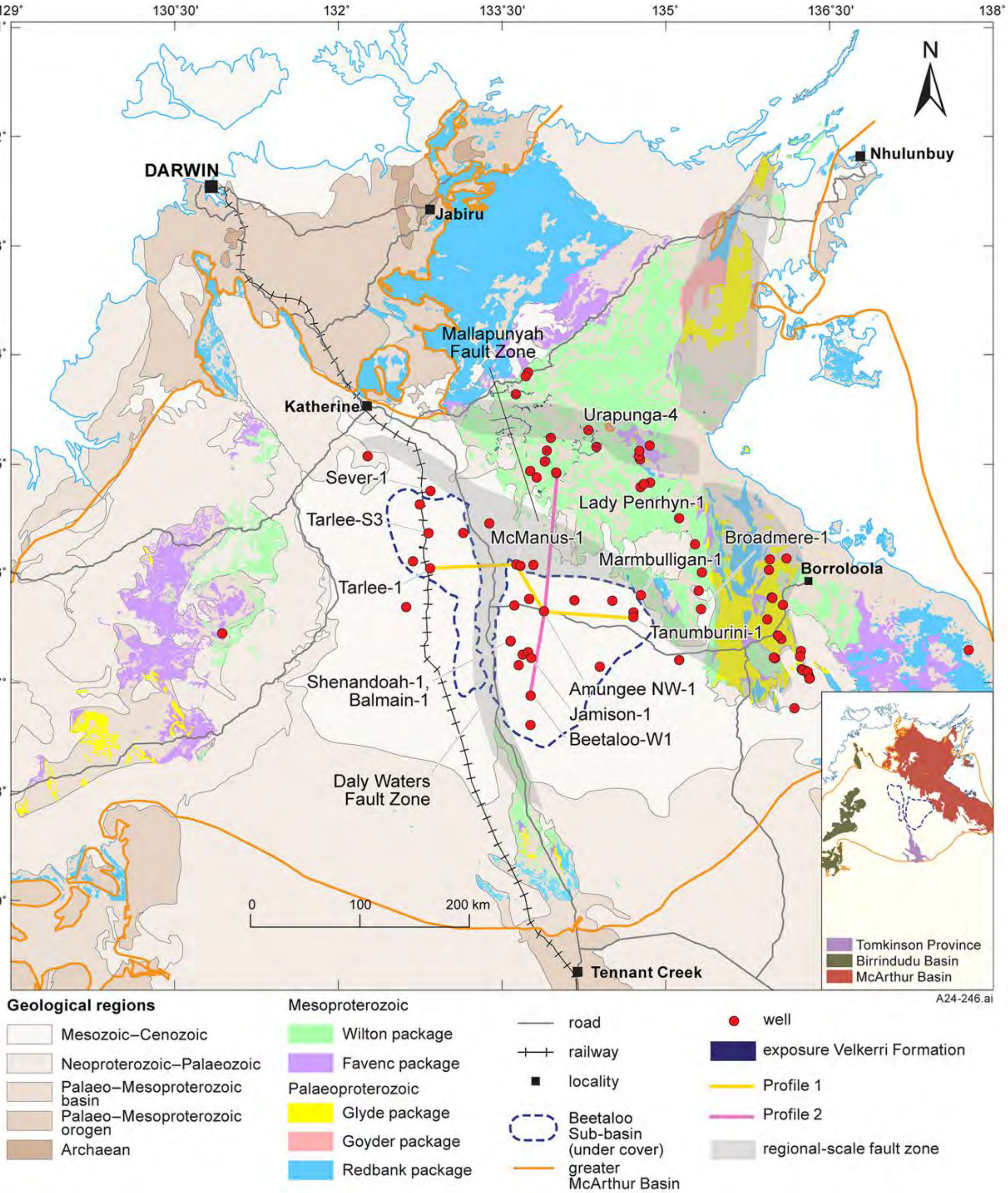

Figure 1. Regional map of the greater McArthur Basin (modified from Jarrett et al., 2022: Figure 2), showing outcrop distribution of Mesoproterozoic and Paleoproterozoic stratigraphic packages in the Northern Territory (NT) (Rawlings, 1999; Close, 2014), and E–W (profile 1) and N–S (profile 2) section lines (Figures 30, 31, respectively). Color coding for packages is the same in Figures 1 and 3. Exploration wells are marked with red circles. Background map is NT geological regions from Northern Territory Geological Survey 1:2.5M geological regions geographic information system data set. Inset map shows distributions of outcropping McArthur and Birrindudu Basins and Tomkinson Province within the greater McArthur Basin.

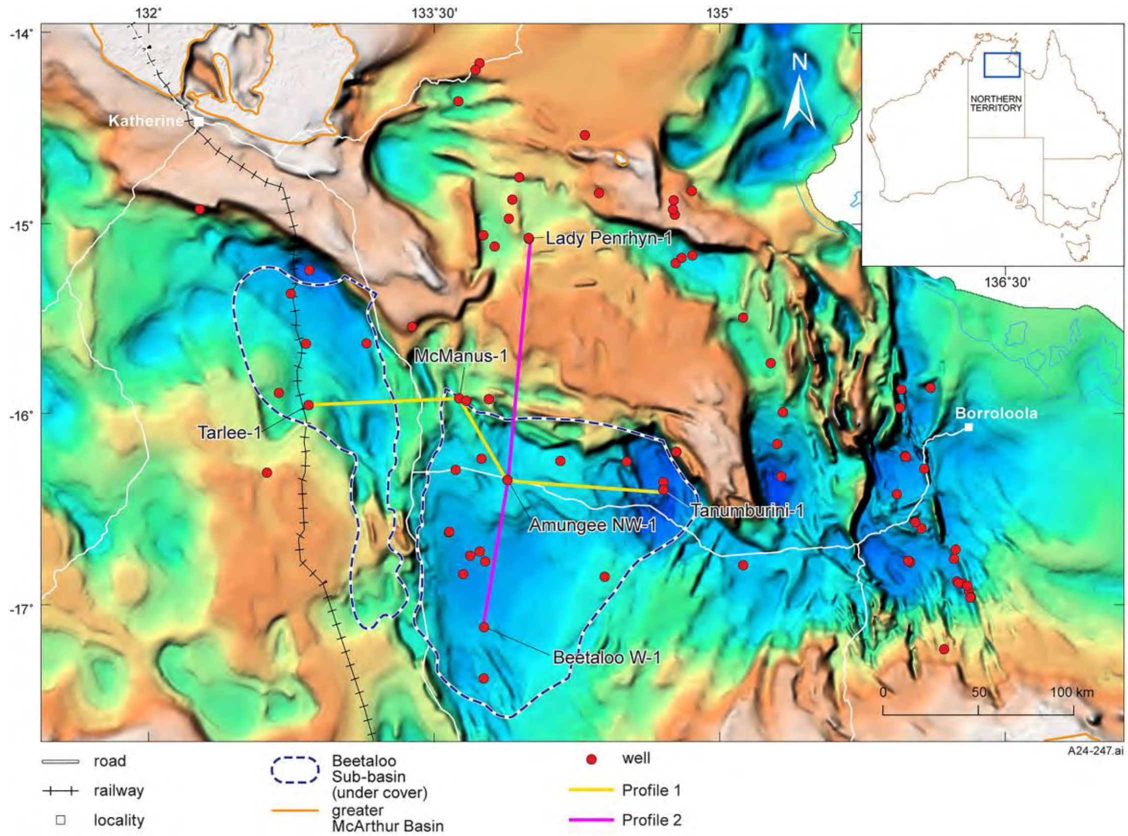

Figure 2. Section lines (profile 1, profile 2) from Figure 1 on SEEBASE® depth-to-basement image (after Northern Territory Geological Survey and Geognostics Australia Pty Ltd, 2021); yellow–brown tones depict interpreted Proterozoic basement highs; blue–green tones are interpreted depocentres.

dynamic subsidence. A period of extension during lower Roper Group times (ca. 1500–1440 Ma) ceased after deposition of this interval, allowing for the great lateral continuity and uniformity of the upper Roper Group, including the Velkerri Formation (Figure 4). This was followed by a post-Roper Group compressional event occurring after 1300 Ma that ended Roper Group sedimentation and resulted in inversion along the DWFZ.

The Velkerri Formation (Sweet in Abbott et al., 2001, after Dunn, 1963) is the focus of this study. In outcrop it is a recessive weathering mudstone-dominated unit with subordinate very fine to fine-grained sandstone, and rare thin dolomitized limestones (Munson, 2016). The Formation is poorly exposed to the north of the Beetaloo Sub-basin and is best known from drillholes within the subbasin (Figure 1). Lanigan et al. (1994) informally subdivided the Velkerri Formation into three intervals (lower, middle, and upper) based on Pacific Oil and Gas Ltd (Pacific) drill intersections in the north of the Beetaloo Sub-basin and adjacent areas. This subdivision has been recognized in most drill intercepts of the unit, including the type section in drillhole Urapunga-4 (Figure 1), and is generally accepted as being characteristic of the Formation.

The three intervals described by Lanigan et al. (1994) have since been formally recognized as the Kalala, Amungee, and Wyworrie Members of the Velkerri Formation, in ascending order (Munson and Revie, 2018; Figure 3). The Kalala and Wyworrie Members have a higher proportion of siltstone (ST) and sandstone relative to mudstone than the Amungee Member, and are generally organically lean. In contrast the Amungee Member mudstones are organic-rich, ranging from 0.3% to 10% total organic carbon (TOC) in the examined cores. Hoffman (2015) noted the presence of highly correlative wireline log markers within the three members of the Velkerri Formation, the most consistent being a thin calcite bed at the base of the Amungee Member, with a characteristic low gamma-ray response

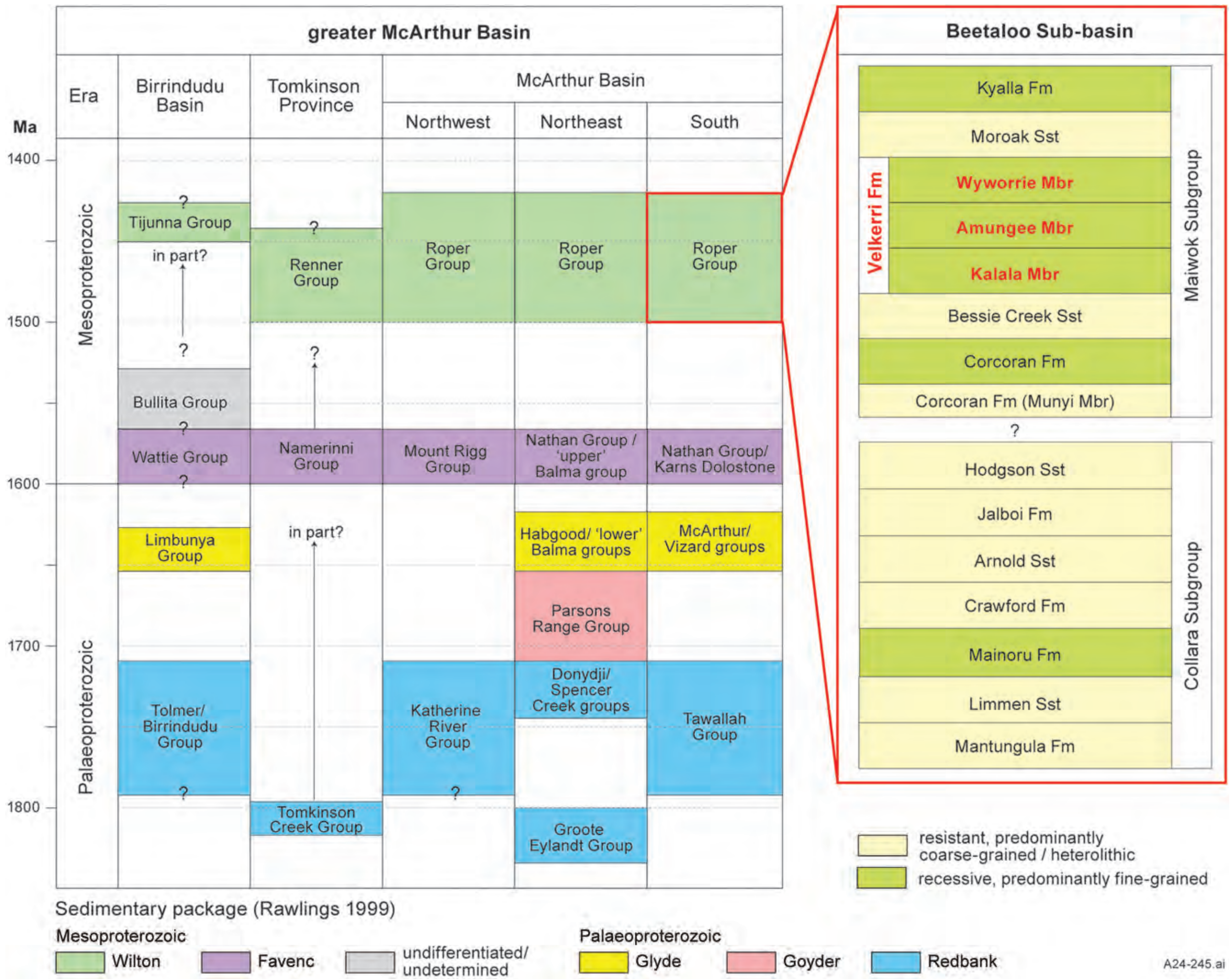

Figure 3. Group level correlation chart for the greater McArthur Basin showing stratigraphic packages of Rawlings (1999) and Close (2014), and Beetaloo Sub-basin Roper Group stratigraphy with Velkerri Formation members highlighted in bold red text (modified from Jarrett et al., 2022). Color coding for packages is same as in Figure 1. Fm = Formation; Mbr = Member; Sst = Sandstone.

followed by a sharp upward step in values (Munson and Revie, 2018), likely caused by an increase in clay minerals and OM. Close et al. (2017) also noted that the Amungee Member is laterally continuous on a regional scale.

Lanigan et al. (1994) described "three broad pulses" of organic rich mudstone deposition in the Amungee Member with TOC mostly ranging from 4% to 7%, and intervening zones of variable organic content. These organic-rich units were informally referred to as units 1–3, in ascending stratigraphic order, by Warren et al. (1998), and later as equivalent units A–C by Hoffman (2015). The three organic-rich shales, particularly the B shale, are of considerable interest to petroleum explorers as a shale gas play because of their high OM content, considerable thickness and lateral extent, and favorable petrophysical characteristics (Côté et al., 2018).

The Velkerri Formation is associated with the Mesoproterozoic Beetaloo Petroleum Supersystem (Jarrett and Munson, 2022; Jarrett et al., 2022), which is the youngest of four petroleum supersystems in the greater McArthur Basin. There are several unconventional shale gas play types within the Beetaloo Supersystem, the most significant of which are the Velkerri shale dry gas and liquids-rich gas plays, both within the Amungee Member (Côté et al., 2018). Organic-rich shales within the Velkerri Formation are thickest and most prospective for unconventional petroleum within and adjacent to the Beetaloo Sub-basin (e.g., Revie, 2017; Cox et al., 2022).

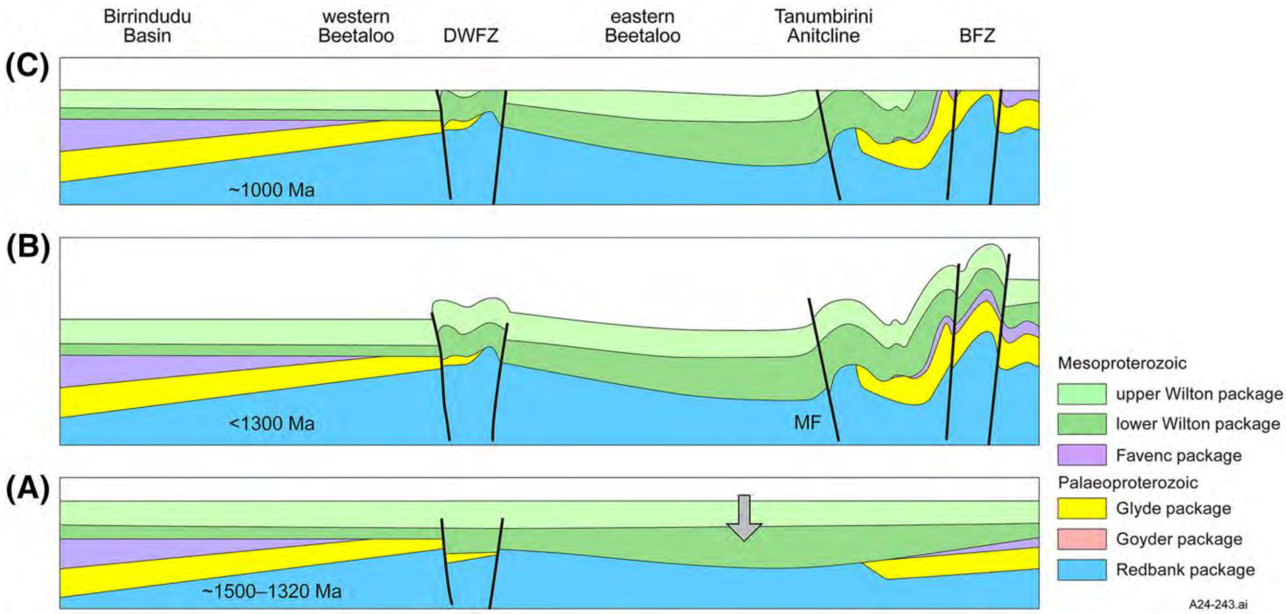

Figure 4. West to east schematic (after Garrad, 2023) illustrating evolution of the Beetaloo Sub-basin from the Mesoproterozoic to the Neoproterozoic. (A) Lower Wilton basin formation in an extensional setting, with a background of dynamic subsidence. Extensional faults were mainly north-south to north-northeast orientation. Fault related subsidence ceased prior to upper Wilton deposition. (B) Compressional deformation–inversion of Daly Waters Fault Zone (DWFZ), Batten Fault Zone (BFZ), early uplift along Marmbulligan Fault (MF). Maximum age constrained by 1330 to 1295 Ma Derim Derim Dolerite (Bodorkos et al., 2021) that predates deformation. (C) Erosion at top Wilton unconformity prior to deposition of Neoproterozoic successions. Gray arrows represent uplift/subsidence. Black lines are faults. The Velkerri Formation and the Amungee Member are part of the Mesoproterozoic upper Wilton depositional "package" (Rawlings, 1999). Color coding of stratigraphic packages is comparable to that used in Figures 1 and 3.

The subject of our case study, the Amungee Member, is host to the world's oldest producible gas shales (Silverman and Ahlbrandt, 2011; Revie, 2017) and has a characteristic "striped" appearance in the studied cores, owing to alternating layers (millimeters to centimeters) of variable TOC content (>4% versus <1%). Additional petrographic elements that are interspersed with the "striped" sedimentological leitmotif are homogenous appearing clay mineral-rich layers and a variety of silt laminae and layers.

Petroleum Exploration History

The McArthur Basin has a long history of exploration for mineral commodities (base metals, uranium, iron ore, manganese, barite, phosphate, and diamonds; Ahmad et al., 2013), conventional oil and gas, and shale gas (Munson, 2014; references therein). Amoco undertook the first major petroleum exploration program in the McArthur Basin that culminated in drilling the Broadmere-1 wildcat well in 1984 (Lanigan et al., 1994), a well that intersected lower to middle Roper Group sediments without significant hydrocarbon shows.

Basin studies by government agencies progressed in the 1980s, including stratigraphic drilling in the Roper Group (Sweet and Jackson, 1986). The upper Roper Group (Figure 3) was identified as having the basin's best source and reservoir characteristics (Jackson and Raiswell, 1991). "Live" oil encountered in stratigraphic drillhole Urapunga-4 also provided evidence of an effective petroleum system.

Pacific Oil and Gas undertook the second major petroleum exploration program in the basin in the years 1986 to 1993, exploring for conventional oil accumulations in the upper Roper Group with 22 exploration wells, and several phases of two-dimensional seismic acquisition (Lanigan et al., 1994). Hydrocarbons shows were recorded in several wells and small volumes of hydrocarbons were recovered from drill-stem tests conducted in the Jamison-1 and Balmain-1 wells in the Beetaloo Sub-basin (Lanigan et al., 1994).

Falcon Oil & Gas Ltd (Falcon) first recognized the potential of Velkerri Formation organic-rich mudstones as a shale gas play. In 2011 Falcon completed one of the earliest hydraulic fracture stimulations of a shale in Australia, successfully flow testing shales in the Shenandoah 1A well (Close et al., 2017). Since then, several operators have been exploring the Beetaloo Sub-basin for shale gas in Velkerri Formation Amungee Member, and the Kyalla Formation. Gas flows rates up to 6.7 MMSCFD (initial production rate over first 90 days [IP90]) have been reported for multistage fracture stimulations of the Amungee Member B shale (Tamboran Resources ASX Release August 11, 2025; ASX stands for Australian Securities Exchange, which publishes company announcements on its web site).

A decade of wildcat drilling by explorers has identified world-class shale gas resources in the Amungee Member, and demonstrated the gas can be successfully extracted at potentially economic rates. At the time of writing explorers were poised to graduate to full appraisal and pilot-production mode (Beetaloo Energy ASX Release September 3, 2025, Tamboran Resources ASX Release September 30, 2025).

Sedimentologic Studies

Santos QNT Pty Ltd (Santos), as operator on behalf of the EP 161 Joint Venture, drilled the Tanumbirini-1 vertical stratigraphic well in 2014 (Figure 1), in order to evaluate the shale gas potential of organic-rich mudstones in the Kyalla and Velkerri Formations (Adderley, 2015) in the deepest buried parts of the Beetaloo Sub-basin, comparatively close to the inferred basin margin (Figure 2). The well terminated at 3950.5 mRT (RT = rotary table, a depth reference) in the Bessie Creek Sandstone after fully intersecting the Kyalla and Velkerri successions, and a 90.18-m 3.5-in. core was acquired over the Amungee Member C shale. In 2016 Santos drilled the Marmbulligan-1 vertical stratigraphic well to evaluate the Amungee Member on the northeastern margin of the Beetaloo Sub-basin, 23 km north-northeast of Tanumbirini-1 (Mackinlay, 2017). The cored (HQ3) well reached a total depth of 675.3 mRT after intersecting a full section of the Amungee Member, documenting the lateral continuity of its organic rich mudstones.

In this study we present detailed sedimentology, facies and diagenesis studies of the Amungee Member in Tanumbirini-1 and Marmbulligan-1, augmented with additional observations made on core material from the distal wells Tarlee-1 and S3 in the northwestern Beetaloo Sub-basin, and McManus-1, and Lady Penrhyn-1 to the north of the subbasin. The results of these studies, contextualized with flume analogue studies that speak to underlying physical, chemical, and biological processes and process parameters, form the basis for this paper. Broadly speaking we recognize two mudstone families in the Amungee, those that were carried into the basin from surrounding areas (extrabasinal silts and muds) and those that reflect processes that dominated the distal parts of the basin (intrabasinal striped shales). The organization of facies descriptions that follows below reflects this perspective.

METHODS

This study is largely based on observations made on six diamond drill cores (Tanumbirini-1, Marmbulligan-1, Tarlee-S3, Tarlee-1, McManus-1, Lady Penrhyn-1; Figure 1) that intercept the Amungee Member, supported by core photographs, thin sections, samples, and analytical data available for the Tanumbirini-1 and Marmbulligan-1 cores. The focus of the study is a sedimentological and petrographic examination of the Tanumbirini-1 core for which the largest amounts of petrographic information were available, and the observations made there are augmented by additional information from the other cores. The Tanumbirini-1 core extends across the C-shale of the member (90 m), the Marmbulligan-1 core (564 m) captures all three subdivisions of the member (A-shale, B-shale, C-shale), and the Tarlee-S3 core mostly captures the B-shale interval (stratigraphic nomenclature follows Crombez et al., 2023).

Initial description of the Tanumbirini-1 core was based on core photographs, followed up by direct core examination at Terratek Labs in Salt Lake City, Utah, that included new core photography on carefully washed core. Understanding the observed lithologies was further supported by examining a set of 70 petrographic thin sections by petrographic microscope (ZEISS Photo III) and scanning electron microscope (SEM), as well as 43 large diameter (12 mm) argon ion milled samples by SEM (FEI Quanta 400 FEG) for insights into early diagenesis and sedimentary fabrics (Schieber, 2013; Schieber et al., 2016). Examination of the Marmbulligan-1 core focused on core photographs and a set of 30 large samples for which polished thin sections and large diameter (12 mm) argon ion milled samples were prepared and studied by petrographic microscope and SEM. For the Tarlee-S3 core, 10 samples were collected and thin sections and large diameter (12 mm) argon ion milled samples were prepared for examination by petrographic microscope and SEM. For McManus-1, Lady Penrhyn-1, and Tarlee-1, core photographs and thin sections housed at the NTGS were studied to better understand the lateral extent of salient sedimentary characteristics.

Thin sections were imaged systematically (bottom to top) at various magnifications and areas of special interest were imaged more extensively. This ensures that final assessments are not biased toward the "highlights" but are anchored in the dominant themes that these samples illustrate. Petrographic imaging was mostly done in plane light, but also with crossed polarizers, and a subset of thin sections was also examined under the SEM for confirmation of critical features in the diagenetic realm.

Thin section examination showed variable carbonate contents of finely disseminated carbonate cement that were largely imperceptible even on well cleaned cores material. Because the available thin sections only cover a tiny amount of the cored intervals (approximately 0.5%), carbonate testing with dilute HCl was conducted at regular intervals (5 cm) to identify intervals of calcareous shale. By testing core material with thin section control, a crude guide to carbonate content was devised, that equated the degree of "fizziness" to carbonate content as seen in thin sections. In addition, rock hardness (a cementation proxy) was evaluated qualitatively with a hardened steel scribe, and quantitatively via Leeb rebound hardness testing (using Proceq SA Bambino 2).

The TOC content and total gamma ray log responses were used in this study to examine basin-wide trends in the Amungee Member stratigraphy. The TOC analyses were conducted by Schlumberger Reservoir Laboratories using LECO TOC for the Tanumbirini-1 core samples (Adderley, 2015), by Energy Resources Consulting using HAWK Pyrolysis for the Marmbulligan-1 core samples (Mackinlay, 2017), and Weatherford Australia using LECO TOC for the Tarlee-1 core samples (Hoffman, 2016). Downhole wireline total gamma-ray logs were acquired by BPB Slimline Services for Lady Penrhyn-1 (Ledlie, 1989), Century Wireline Services for McManus-1 (Lanigan and Ledlie, 1990), and Schlumberger for Amungee NW-1 (Origin Energy Resources, 2016), Beetaloo W-1 (Origin Energy Resources, 2017), Tarlee-1 (Hoffman, 2016), and Tanumbirini-1 (Adderley, 2015). Total gamma-ray logs for the wells noted in Figure 1 were used for cross-basin stratigraphic correlations, adopting gamma-ray log picks by Crombez et al. (2023).

DESCRIPTION OF SHALE FACIES

Describing the Amungee Member may seem at first glance a simple matter, and 20 yr ago it would have been perfectly acceptable to describe the bulk of it as "dark gray siltstone grading to claystone" and be done with it. Notwithstanding, the fact that many mudstone successions are heterogeneous and sedimentologically variable at the submillimeter-to-decimeter scale vertically and at the centimeter-to-kilometer scale laterally has been explored for some time (Schieber, 1989, 1990; O'Brien and Slatt, 1990; Macquaker et al., 1996, 1998; Bohacs, 1998; Guthrie and Bohacs, 2009; Lazar and Schieber, 2022), and the concepts developed in those publications (Lazar et al., 2015a, b) can be employed quite profitably to the study of the Amungee Member (Figure 5).

Because the terms shale and mudstone are both in common use for the description of fine grained sediments, our convention for the Amungee cores is as follows: We consider the Amungee Member as a shale stratum, and within it the term shale is used for those rocks that show sub-millimeter to millimeter-scale layering, and the term mudstone is applied to multimillimeter-to-centimeter scale layers that appear uniform and lack discernible internal structure (Figure 6). The three elements of a comprehensive shale description are (1) fabric and textural elements (sedimentary structures, fine scale layering), (2) grain size, and (3) composition (e.g., Lazar et al., 2015a, b). Yet, although textural elements and grain size are approachable via visual inspection and the use of a hand lens, getting a measure of composition is not as simple. Composition can be assessed via information from thin section study, or from x-ray diffraction and whole rock chemistry data, neither of which tends to be available to the practicing geologist at the time the core (or outcrop) is being logged. Thus, as a matter of practicality, initial

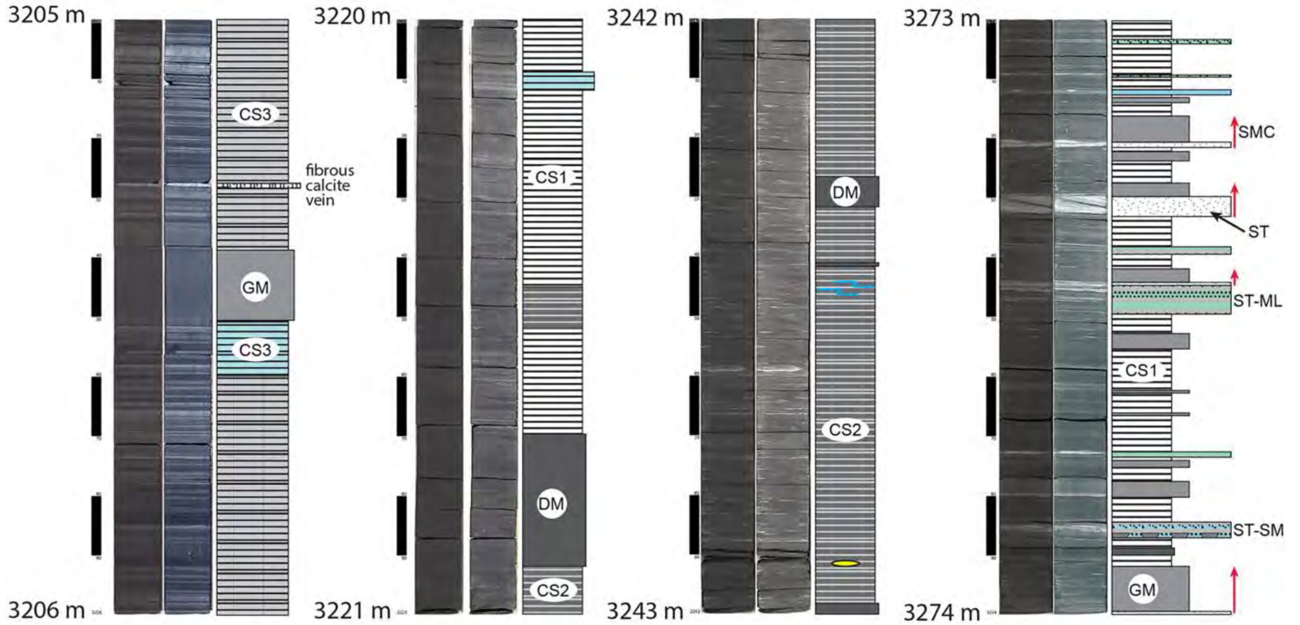

Figure 5. Four lithologic logs that illustrate facies variability and heterogeneity in the Tanumbirini-1 core that was recovered from the uppermost subunit (C-shale; Crombez et al., 2023) of the Amungee Member (the core spans an interval of 90 m). The letter codes (SMC, GM, etc.) and stratigraphic column fill patterns are keyed to the facies overview in Figure 6 and a detailed stratigraphic section in the paper's online repository (Online Supplement 1, available as AAPG Datashare 216 at www.aapg.org/datashare). Lithologic logs and facies are closely comparable for the Marmbulligan-1 core, whereas the Tarlee-S3 core is dominated by distal facies faintly striped carbonaceous shale (CS2) and dark mudstone (DM) (Figure 6). GM = gray mudstone; ML = muddy laminae; SM = soupy mud; SMC = silt-mud couplet; ST = siltstone.

facies categorization relies heavily on the "visual" (texture, grain size), and is then augmented (compositional detail, etc.) as more information becomes available subsequently.

On the basis of core photographs, careful core examination and a study of more than 100 thin sections, the shales of the Amungee Member can clearly be differentiated, but subtlety abounds. In essence, we have a "background" of dominantly black shale with a "striped" character (laminated to thinly bedded), that is interspersed with event beds that are light gray in color and may have a silty base. This striped shale "background" constitutes most of the strata in the Amungee Member and has been subdivided into four genetically intertwined (interleaved/interbedded) facies subtypes (Figure 6). In addition, there are beds that are more intensely cemented (by silica and Fe-carbonates) and these may stand out due to hardness and lighter color. Figure 6 summarizes the shale facies types that are described in the following paragraphs. Because shale successions typically exhibit much larger variations in texture and fabric than in composition (Schieber, 1989, 1999a), differentiation of shale facies is focused on characteristics of texture and fabric that reflect sedimentary and diagenetic processes. Building on the classification philosophy by Lazar et al. (2015a), the approach taken here integrates core specimen and thin section-scale palaeontologic, sedimentologic, and petrographic observations, extending the microfacies concept of Brown (1943), Cuvillier (1952), and Fluegel (2004) to shales and mudstones. Our practice of "shale microfacies analysis" is also summarized in Ulmer-Scholle et al. (2015), and we use the term shale as a root name for rocks with visible lamination (not to be confused with fissility that results from weathering), whereas mudstone denotes rocks that appear homogenous and unlaminated. Because pictures provide the most direct access to the multifaceted entities that are "shale" and convey the circumstances of their origin in the most compact way possible, we heavily rely on visual representation to decode the sedimentary origins of the Amungee Member shale succession.

Silt and Mudstone Layers

Silt and silt-bearing layers are a minor lithology in the Amungee Member, but are nonetheless important.

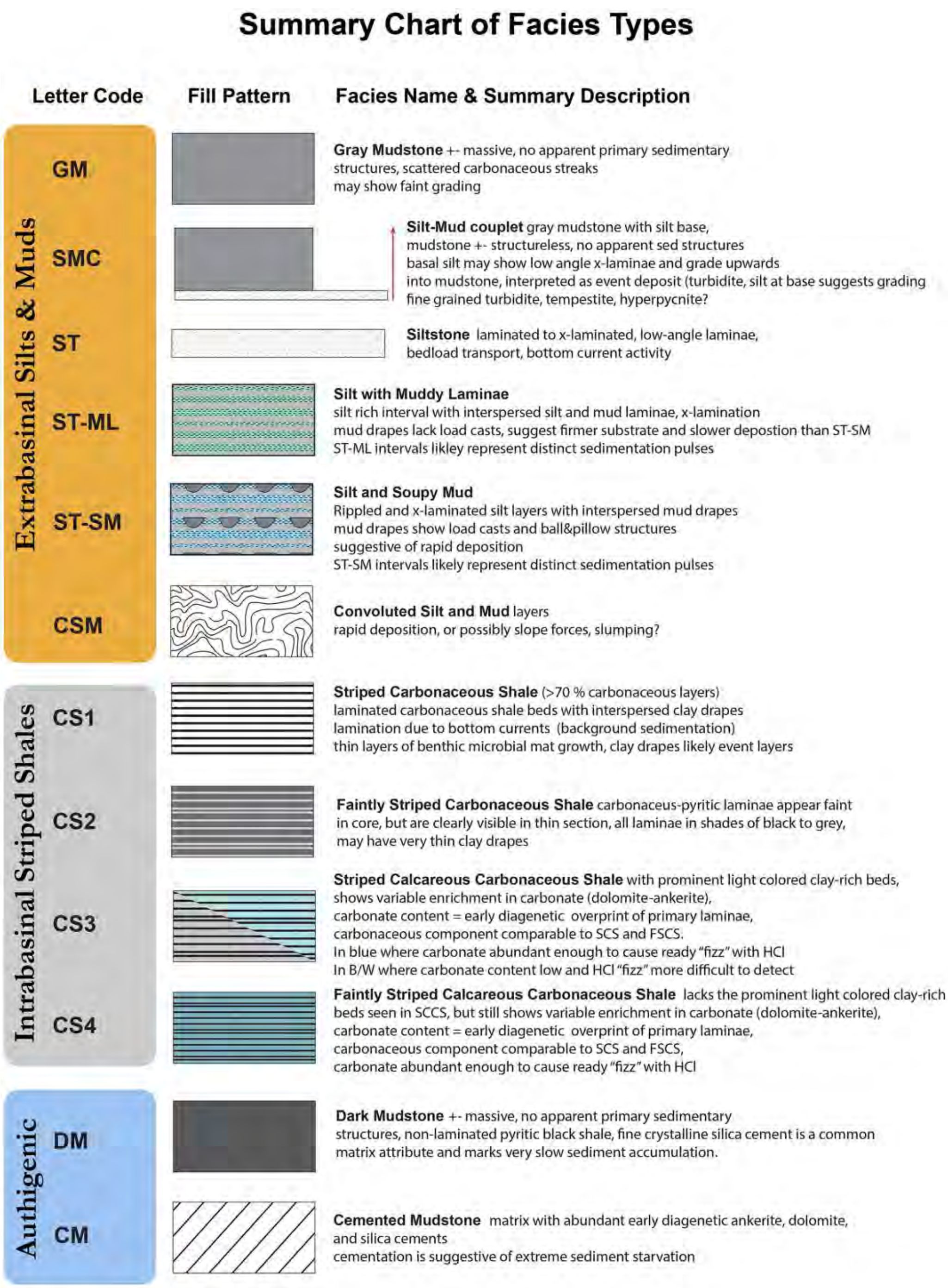

Figure 6. Facies types observed in the Amungee Member of the Velkerri Formation. The symbols, fill patterns, and facies designations are used in Figure 5 and in the detailed stratigraphic section (Online Supplement 1, available as AAPG Datashare 216 at www.aapg.org/datashare).

Gray mudstone (GM) is grouped with silty layers because they are an integral part of their deposition. Silt-rich layers, also called coarse mudstones (Lazar et al., 2015a, b), are more readily visible in core than most other lithologies (Figure 6), but are a complex and mingled lot when examined in detail, and as a consequence five different types were recognized and logged.

GM: Gray Mudstone Beds, and SMC: Silt-Mud Couplets

Considered cogenetic, GM and silt-mud couplets (SMC) are described together. The GM beds typically range in thickness from 5 mm to a few decimeters, but may reach as much as 0.8 m (Figure 7A; Online Supplement 1, available as AAPG Datashare 216 at www.aapg.org/datashare), and are associated with SMC beds of comparable thickness (may reach 0.45 m). The SMC beds are in essence GM beds with a basal silt-rich part that shows lamination, cross-lamination, soft-sediment deformation, and gradational tops (Figure 7B). Basal cross-laminae indicate bedload transport of silt, and soft sediment deformation (Figure 7D, E) suggests fabric collapse due to water escape, likely related to rapid deposition. The GM and SMC beds can show soft-sediment deformation for part or all of their thickness (described as convolute silt/mud [CSM], Figure 6).

Silty bases of SMC beds can thin to the point of becoming invisible in core, yet in thin section they may still be discernible (Figure 8). Thus, although GM and SMC beds are differentiated according to what is visible in clean cores, they plausibly record

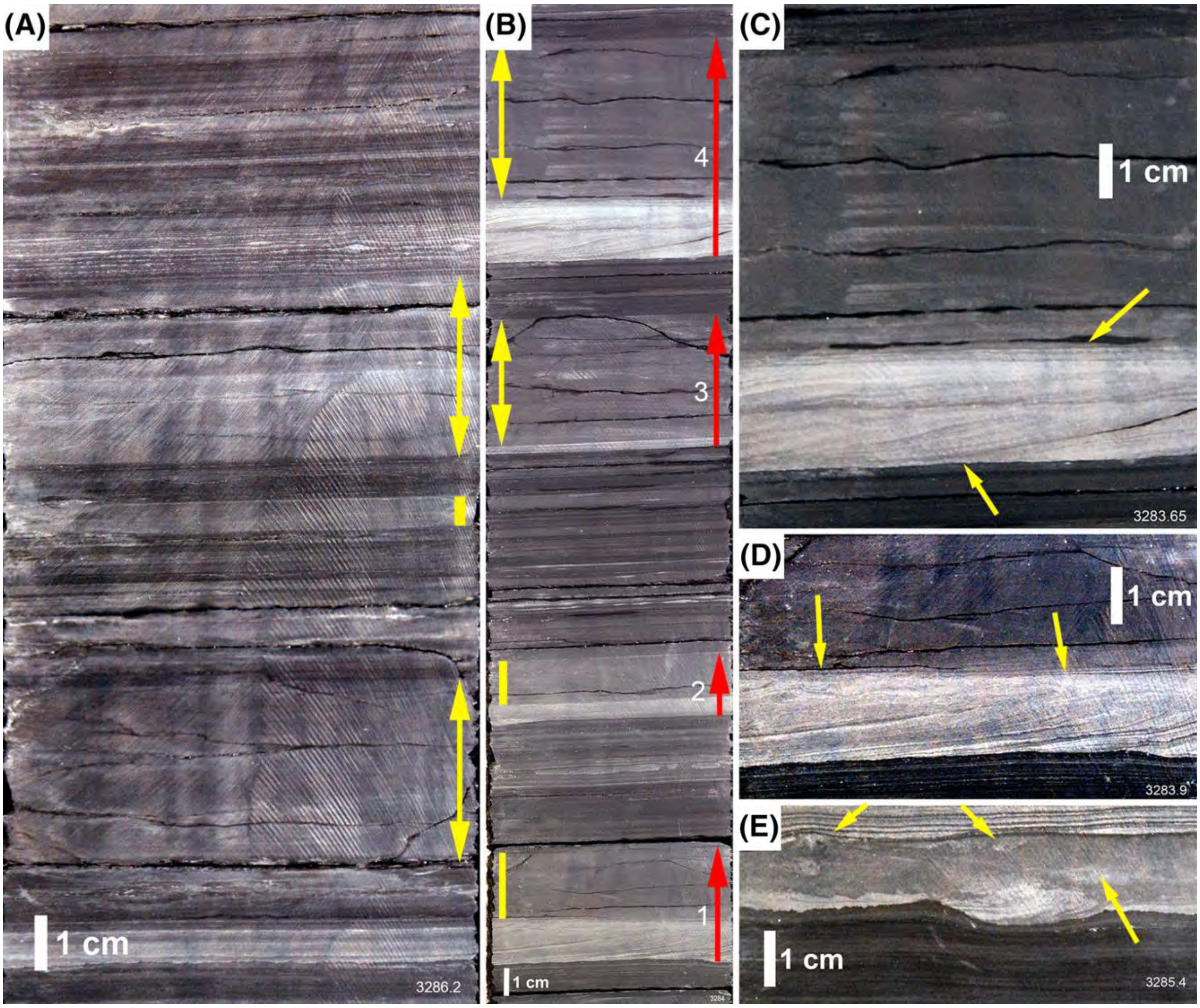

Figure 7. Images of facies GM (gray mudstone) and SMC (silt-mud couplets). (A) Core that shows three examples of GM beds (yellow double arrows and yellow bar on right) that are mainly interbedded with darker colored striped shale. The GM beds appear massive and structureless. (B) Core section that shows multiple SMC beds (numbered and marked with red arrows) that are interbedded with darker colored striped shale. The GM parts are marked with yellow double arrows and bars on left. (C) Enlarged view of SMC bed 4 from (B). Basal silt (yellow arrows) consists of two stacked cross-laminated intervals. (D) SMC where basal silt is cross-laminated and shows soft-sediment deformation near the top (yellow arrows). (E) SMC with sunk-in cross-laminated ripples at base, and internal soft-sediment deformation (yellow arrows). Note that GM beds and the GM parts of SMC beds commonly show irregular subhorizontal fractures when compared to intervening intervals of carbonaceous (striped) shale.

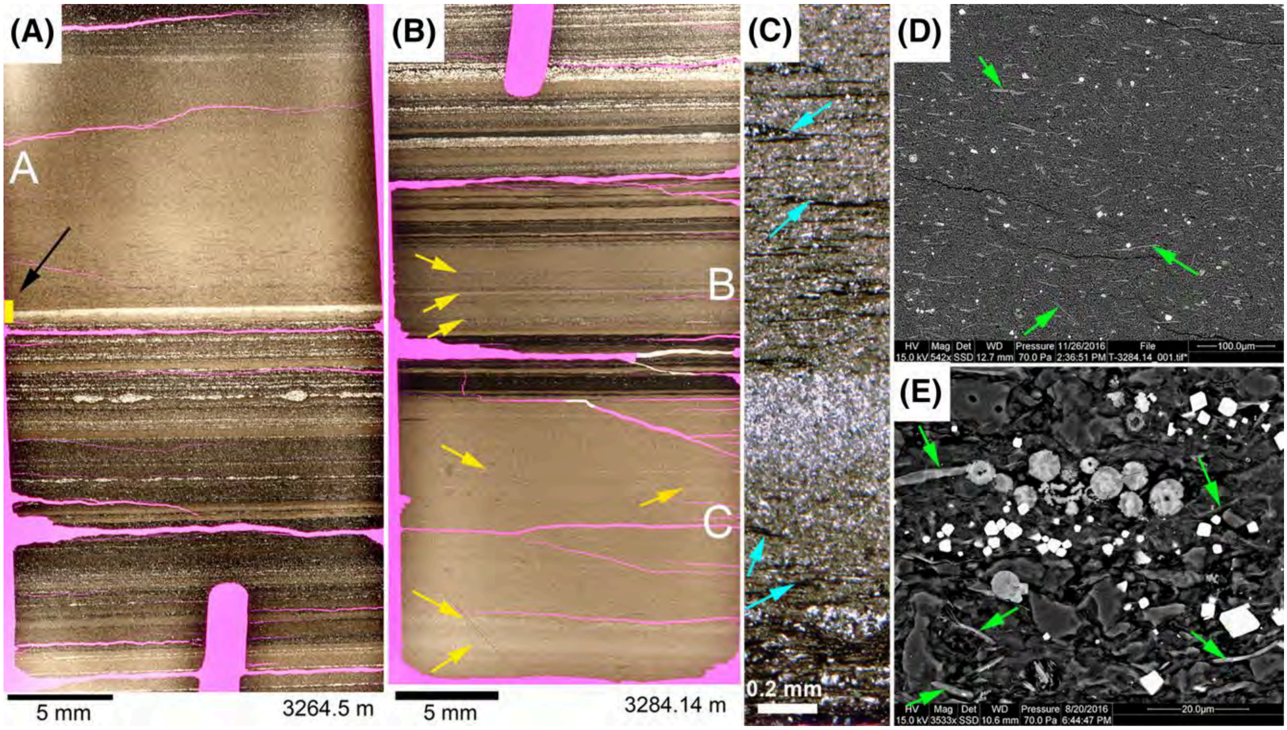

Figure 8. (A–C) GM (gray mudstone) beds in thin section. (A) Layer A shows GM bed with sub-millimeter silt layer near base that has load casts into underlying GM. Arrow points to yellow bar that locates the enlarged view in (C). (B) Details of GM layers (marked B, C). Yellow arrows point to faint internal lamination. (C) Detail from GM layer in A (yellow bar). The GM contains carbonaceous streaks (blue arrows), up to several millimeters long and variably deformed, likely eroded from benthic microbial mats (Schieber, 1989). (D) Scanning electron microscope (SEM) image (backscatter) of typical GM matrix. Abundant chlorite (green arrows) and mica flakes show a crude horizontal alignment; other flakes that are near vertical and subvertical hint at an originally randomized-flocculated fabric. (E) For comparison, SEM image (backscatter) of carbonaceous lamina in striped shale which also shows abundant chlorite flakes (green arrows), that are smaller in size than those seen in GM beds.

the same depositional process parameters, with the “outcome” dependent on how much silt is available to form the basal deposit.

For the sake of core logging efficiency, GM layers less than 5 mm thick were described as “clay drapes” and those thicker than 5 mm as GM beds, with the implicit understanding that both types of layers were genetic beds that reflected comparable conditions. On the basis of SEM examination, it is apparent that GM and SMC layers, as well as clay drapes and carbonaceous layers of the striped shales (facies CS), contain the same assemblage of terrigenous components (detrital quartz, feldspars, phyllosilicates, micas, and chlorite) and have a common detrital heritage (Figure 8D, E).

Grading at various levels of subtlety is one of the main sedimentary features of GM and SCS beds. They contrast sharply with intervening, much more thinly bedded/laminated sediments (striped carbonaceous shale [CS1]). Diagenetic dolomite and ankerite (up to 100 μm) crystals may be scattered through the clay-rich matrix of GM and SMC beds, as well as micron to nanometer sized diagenetic quartz. Appearance and textural context of these cement minerals is the same as seen in striped shales (described below), but overall abundance is notably less. The inference that these rocks should be comparatively weak mechanically is supported by the ease of scratching it with a metal scribe, low Leeb rebound hardness, and the fact that they fracture more readily (see Figure 7).

ST: Siltstone, ST-ML: Silt with Muddy Laminae, and ST-SM: Silt and Soupy Mud

Silt not only occurs in SMC beds. Discrete silt-rich layers with a range of sedimentary features are common in the Amungee and are in most cases intermingled with clay drapes (composition like GM) and carbonaceous (striped) shale. The challenge for the practicing geologist is how to balance a desire for accuracy with logging efficiency, a problem we approached as follows.

If a silty interval contained more than two-thirds silt it was categorized as ST (just silt), and if there was more than one-third of clay-rich laminae, it was called "silt with muddy laminae" (ST-ML). If there was fabric collapse of associated muds (large load casts, ball and pillow), it was logged as "silt and soupy mud" (ST-SM), because these features typically form in water-rich muds. These various types of silt layer are illustrated in Online Supplements 2–4, available as AAPG Datashare 216 at www.aapg.org/datashare. See Online Supplement 1 (available as AAPG Datashare 216) for stratigraphic distribution.

Compositionally all silt layers described here consist of a mixture of detritus composed of quartz silt and a substantial proportion of chlorite. Under the SEM, due to differences in backscatter coefficients, the ST layers have a dark (quartz) versus light (chlorite) speckled appearance (Figure 9A–C). Additional common detrital silt grains are feldspar and muscovite, as well as scattered mudstone lithics (Figure 9C).

CSM: Convoluted Silt and Mud Layers

Soft-sediment deformation is observed in various parts of the Tanumbirini-1 and Marmbulligan-1 cores. The most readily recognized type of soft-sediment deformation is the CSM facies (Online Supplement 4A, available as AAPG Datashare 216

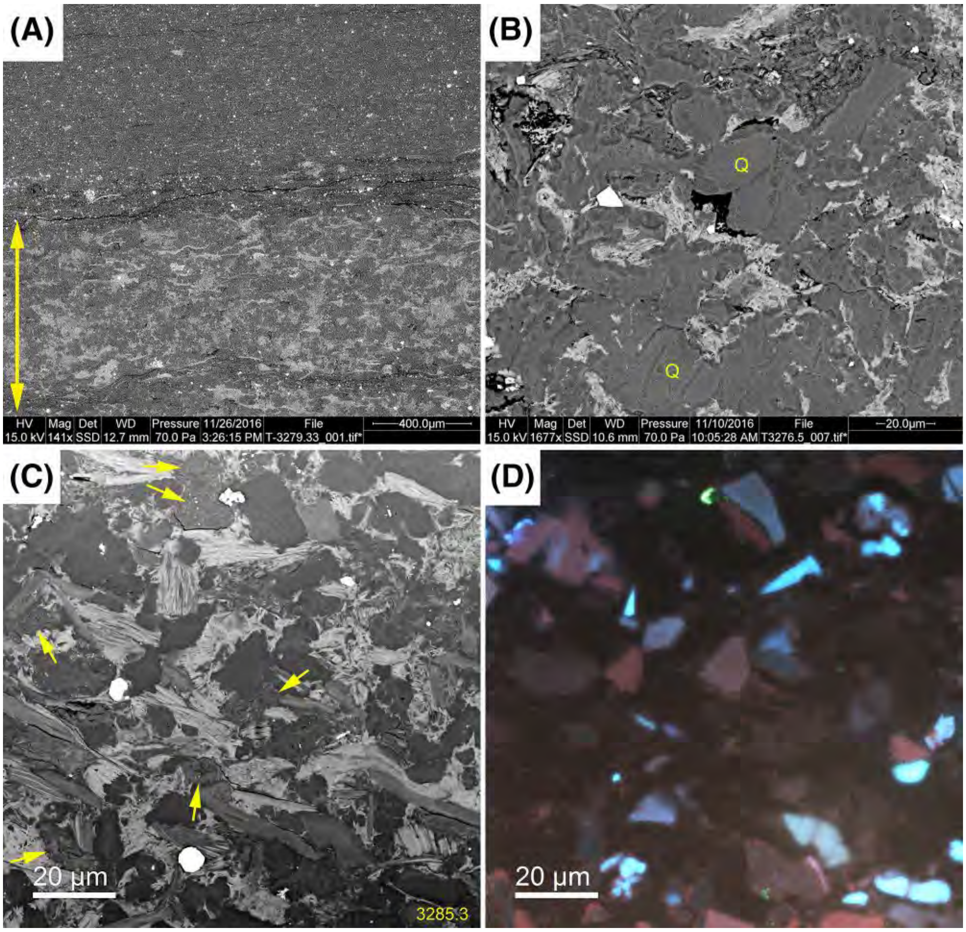

Figure 9. SEM–based (scanning electron microscope) petrographic observations of silt layers. (A) Silt layer (yellow double arrow) shows a speckle pattern of dark and light spots. (B) A closer view that shows that the speckle pattern largely reflects a mixture of quartz grains (Q) and detrital chlorite (bright and variably deformed). (C) A similar image as B, but ion milled surface shows more clearly defined quartz grains (dark gray), muscovite (medium gray flakes), abundant chlorite (bright) flakes, and even some mudstone lithics (yellow arrows). (D) SEM CL (cathodoluminescence) image of same area shown in (C). The image shows abundant dull colored, moderate CL detrital quartz grains (bright blue is K-feldspar). Reddish grains are from low-grade metamorphic source, dull bluish-brownish grains are from medium-grade metamorphic source rocks (Zinkernagel, 1978; Schieber and Wintsch, 2005; Schieber, 2016b; Wilson and Schieber, 2024).

at www.aapg.org/datashare) which is associated with GM and SMC beds (see Online Supplement 1, available as AAPG Datashare 216 at www.aapg.org/datashare, for stratigraphic distribution).

The "Striped Shales"

This facies type represents the lithologic essence of the Amungee Member, and its subtypes (Figure 6) share multiple salient characteristics that relate to a common origin. These characteristics are introduced with the description of the most common type, CS1, and then will be documented only briefly for the remaining striped shale subtypes.

CS1: "Striped" Carbonaceous Shale

Together with facies "faintly striped" carbonaceous shale (CS2), this is the most common rock type in the Tanumbirini-1 and Marmbulligan-1 cores. The name is "literal" and descriptive, and reflects (Figure 5) the striped nature imparted by millimeter-thick clay drapes that alternate with layers of carbonaceous shale. It is a common shale facies in Proterozoic shale successions elsewhere (e.g., Schieber, 1986, 1989, 2007a).

The thin section scans in Figure 10 illustrate commonalities as well as variability from layer to layer. Namely, the ubiquity of clay drapes (light brown) that range from a fraction of a millimeter to several millimeters in thickness, and are interspersed with carbonaceous layers (dark) that are of more than one type. For example, the thick carbonaceous interval in the upper half of Figure 10B has even parallel thin silt laminae, whereas other carbonaceous layers appear homogenous and vary in darkness (organic content). The building blocks of facies CS1 are described and illustrated in the Carbonaceous Layers and Clay Drapes sections below.

Carbonaceous Layers—The carbonaceous layers that dominate facies CS1 display three different microfabrics—parallel laminated, wavy-anastomosing, and homogenous-lenticular. These can be interlaminated at the submillimeter to millimeter scale, and identification and differentiation is only possible in thin section.

"Parallel laminated" is expressed as thin parallel coarse silt laminae separated by finer material. Silt laminae show lateral thickness variability and may show thickening lenses suggestive of microripples (Figure 11; Yawar and Schieber, 2017).

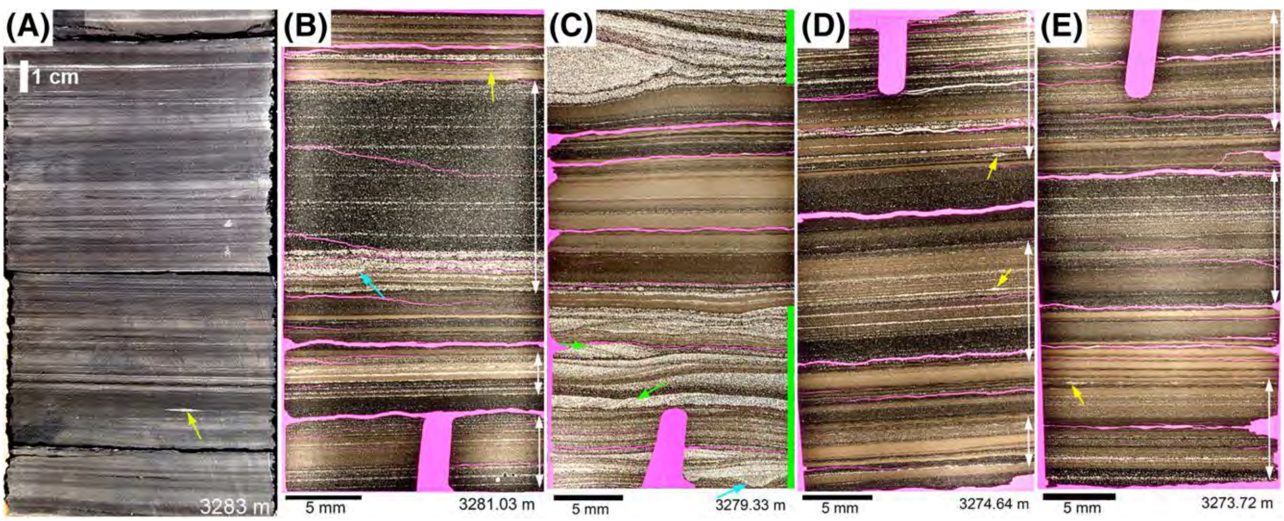

Figure 10. (A) Core photograph (washed, contrast enhanced) from the Tanumbirini-1 core, which is representative of "striped" carbonaceous shale (CS1). Yellow arrow points to silt lens. (B–E) Thin section scans that show more closely the alternating organic poor layers (light brown) and organic-rich layers (darker) that produce the striped appearance of these rocks. White double arrows indicate "parallel laminated" carbonaceous intervals with thin silt laminae (see below). (C) Image shows an interval of CS1 sandwiched between intervals of "silt with muddy laminae" (ST-ML; marked with green side bar). Yellow arrows point to small silt lenses, blue arrows point to local scours, and green arrows point to examples of cross-lamination in ST-ML. The pink areas in the thin sections are due to stained epoxy resin. Although it is common practice to mark "stratigraphic up" in thin sections with a saw-cut notch, sedimentary features indicate that thin sections (B) and (C) were marked the wrong way. They were therefore rotated by 180°, as were other mismarked thin sections pictured in this paper. This is a problem common to many collections of shale thin sections.

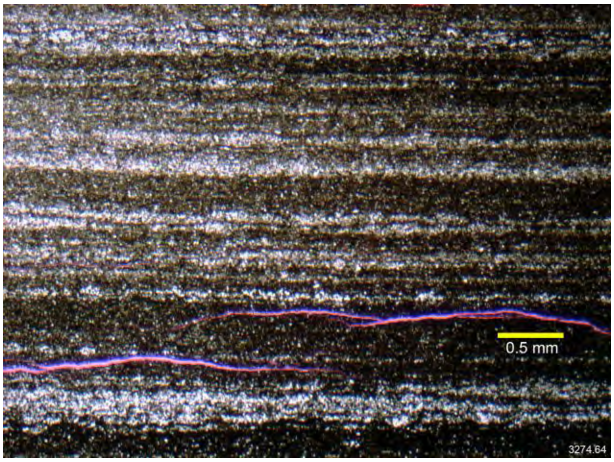

Figure 11. Parallel coarse silt laminae in carbonaceous layer of CS1 (striped carbonaceous shale) (Tanumbirini-1). Variable thickness and lenses suggest that silty laminae were deposited by bottom currents (Schieber, 2011; Yawar and Schieber, 2017).

"Wavy-anastomosing" carbonaceous laminae as seen in Figure 12 are difficult to examine in thin sections of standard thickness (30 μm) as they are largely opaque in transmitted light (Figure 9A). Examined in ultrathin sections (20 μm or less), they consist of continuous wavy carbonaceous laminae (Figure 12C) that contain interspersed sediment as discontinuous thin laminae and lenses. When there is enough sediment supply, these carbonaceous laminae are more apparent because they are fully separated by thin clay drapes (Figure 12B). In the literature, this texture has been attributed to benthic microbial mats (e.g., Schieber, 1986).

"Homogenous-lenticular" carbonaceous laminae have a textural kinship to flaser bedding and are characterized by layers of tightly packed lenses of variable composition (Figure 13). The particulate nature of these lenses is quite obvious in argon ion-milled images (Figure 14). Clusters of micron-size phosphate spheres are commonly associated with carbonaceous material (Figure 14D).

Given that these lenses appear to be compacted fragments of a preexisting muddy substrate (Figure 14), it is interesting to note that although they may occur randomly mixed within a given layer, specific types of lenticles can strongly dominate thin intervals (Figure 15) within them. In a number of instances, thin sections also showed an inclined-imbricated fabric within millimeter-thick lenticle layers, suggestive of bedload transport of original particles produced by bottom current erosion of contemporaneous seafloor muds (Figure 16; e.g., Schieber et al., 2019).

Of the three types of carbonaceous laminae that are observed in the Amungee Member, the homogenous-lenticular type strongly dominates in all striped shale subtypes (Figure 6). Based on the examined thin sections it characterizes approximately 70%–80% of the carbonaceous layers, followed by the parallel (silt) laminated variety (approximately 20%), and the wavy anastomosing type (<5%–10%). The carbonaceous layers that separate silt laminae (Figures 10B, 11) tend to be of the homogenous-lenticular type as well.

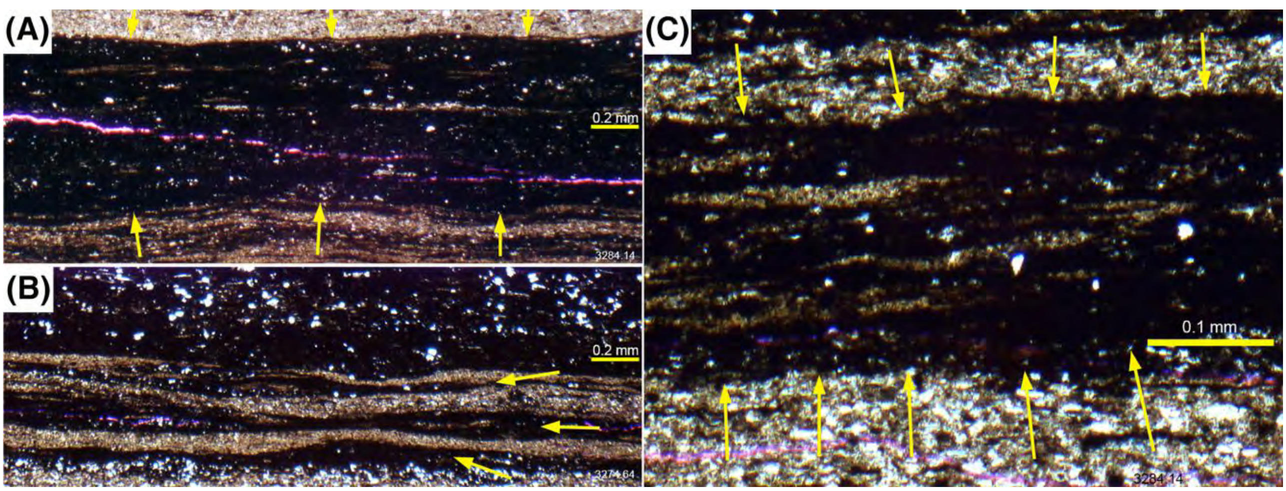

Figure 12. Photomicrographs showing closeup views of wavy-anastomosing carbonaceous laminae (Tanumbirini-1). (A) Set of wavy anastomosing carbonaceous laminae between yellow arrows. (B) Continuous carbonaceous laminae (yellow arrows) with intervening clay drapes. (C) Set of carbonaceous laminae (between yellow arrows) with interspersed discontinuous clay drapes and lenses.

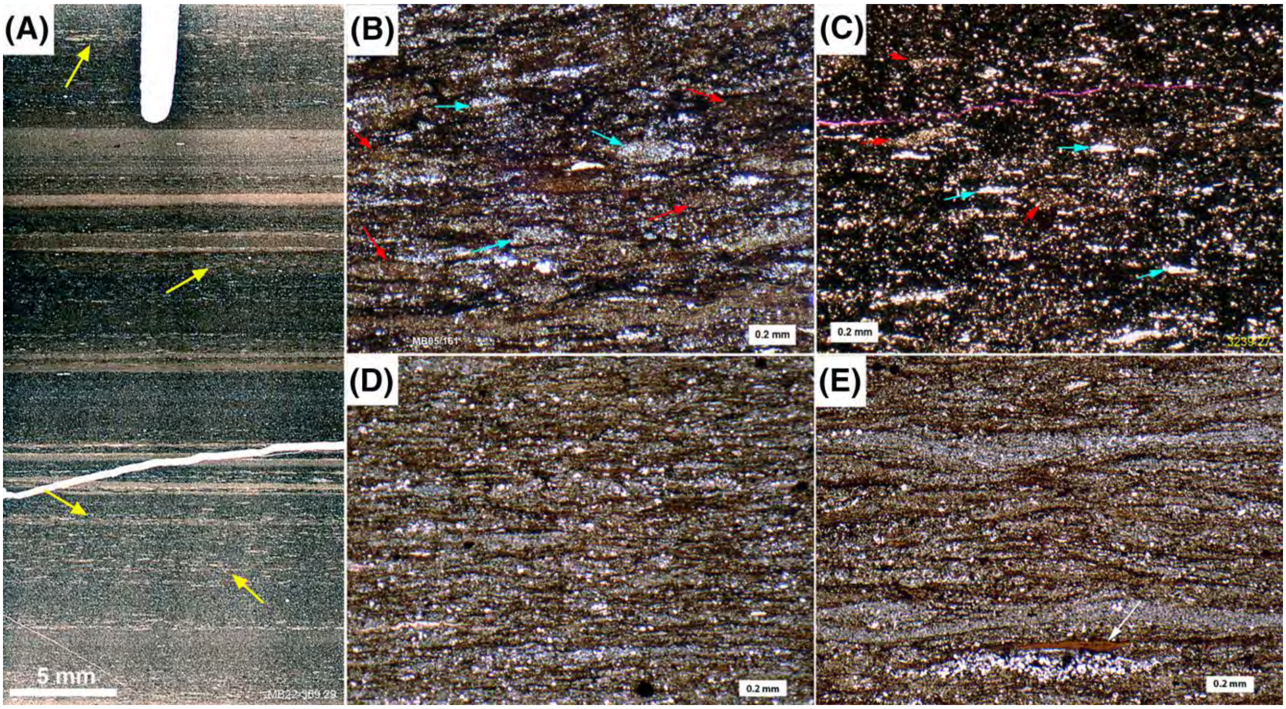

Figure 13. Petrographic expression of homogenous-lenticular carbonaceous laminae (Tanumbirini-1, Marmbulligan-1). (A) Thin section scan, several of the dark layers show light lenses in a darker matrix (yellow arrows). (B) Abundant light-colored lenses in darker matrix. (C) An example with a higher proportion of dark matrix. Different lenticle types pointed out by red (brown lenticles) and blue (light lenticles) arrows. (D, E) Illustration of size range of lenticles. (D) Small lenticles of 0.1 to 0.5 mm length. (E) Large lenticles of 0.5 to 1 mm length. White arrow points to compressed organic cyst.

Clay Drapes—Submillimeter to millimeter-thick clay drapes (Figure 17) that are interspersed with above-described carbonaceous laminae contrast strongly by being largely free of carbonaceous/OM. The clay drapes are basically of the same composition as the GMs of facies GM and the upper parts of SMC (facies SMC, Figure 6), an attribute suggestive of genetic linkage (see descriptions and further discussion below). They may in places show basal silt, grading and micro load casts into underlying laminae, and lack the phosphate spheres (Figure 14D) that are ubiquitous in carbonaceous laminae.

CS2: "Faintly Striped" Carbonaceous Shale

This facies type shares multiple characteristics with CS1, such as clay drapes, variably light layers with rip-up material, carbonaceous layers with parallel lamination, wavy-anastomosing layers, and homogenous-lenticular layers. At its simplest, it can be considered a CS1 variant that has fewer and thinner clay drapes interbedded with carbonaceous shale laminae and as a result appears more featureless when compared to CS1. Strongly contrast-enhanced images show well-defined visible layering (Figure 18A, B) that what would likely be reported as faint lamination during a routine core description.

Layers dominated by lenticular fabric (Figure 15) form the bulk of CS2. They vary in color from dark to medium gray, and form millimeter to tens of millimeters-thick layers. In addition, the rock may contain streaks (same dimension as lenticles) of diagenetic carbonate (ankerite) and silica (quartz). Lenticles (rip-ups) typically are 100 to several hundred microns in length, but may reach more than 1 mm in certain intervals (Online Supplement 5, available as AAPG Datashare 216 at www.aapg.org/datashare).

Wavy-anastomosing layers, plausibly associated with benthic microbial mats (Schieber, 1986), occur as thin intervals (Online Supplement 5B, available as AAPG Datashare 216 at www.aapg.org/datashare), but are very similar in appearance to compacted rip-up clasts. Presumed benthic microbial mats may represent only a small part of facies CS2.

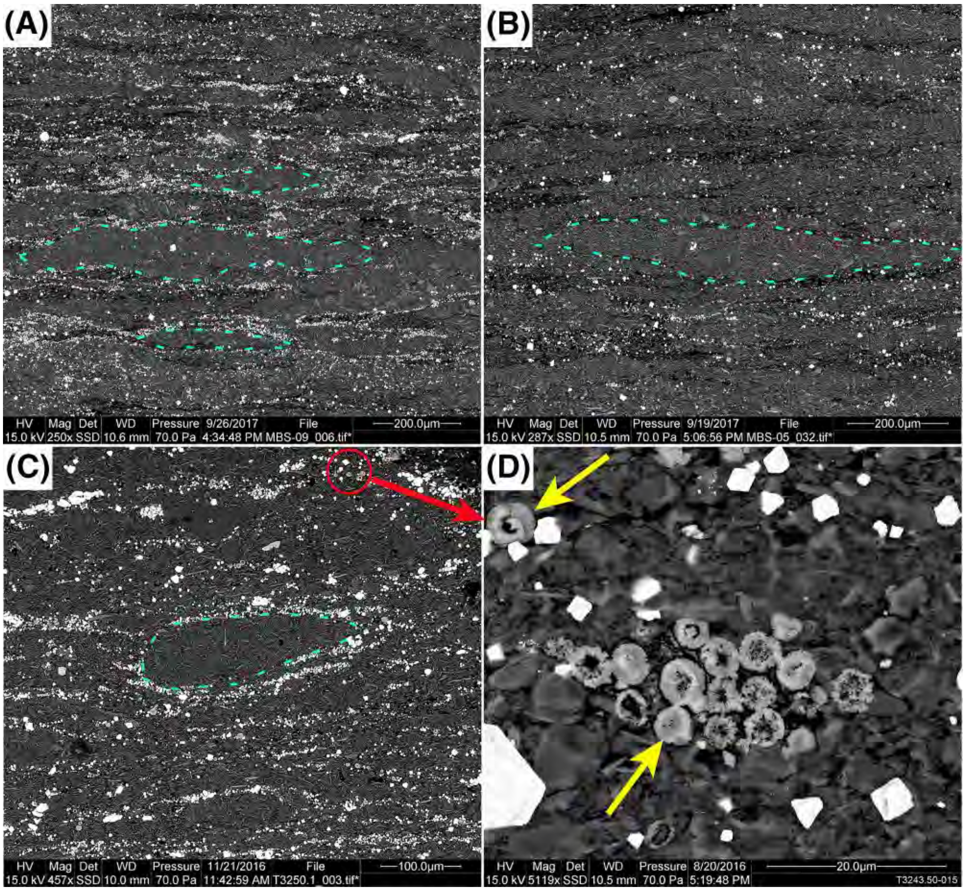

Figure 14. (A–C) SEM (scanning electron microscope) backscatter images from homogenous-lenticular carbonaceous laminae show petrographically distinct deformed-flattened lenticles (some examples outlined with dashed blue line) that are surrounded-bordered by a pyrite-rich (bright grains) matrix. The red circle marks typical matrix, and (D) shows matrix composition and texture at higher magnification with dispersed organic matter (black), pyrite grains (bright), and scattered and clustered phosphate spheres (yellow arrows). (A) and (B) are from Marmbulligan-1, (C) and (D) are from Tanumbrini-1.

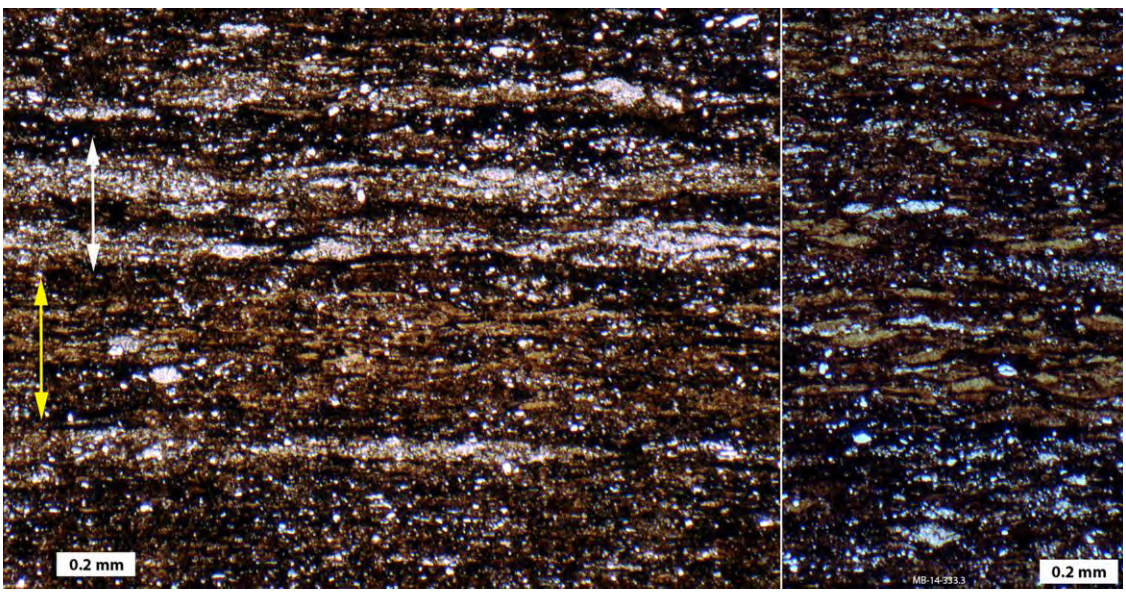

Figure 15. Photomicrograph showing layer by layer variations in lenticle type (Marmbulligan-1), an indication that through time, lenticles were sourced from different muddy substrates. Left image is plane light, right image is crossed polarizers.

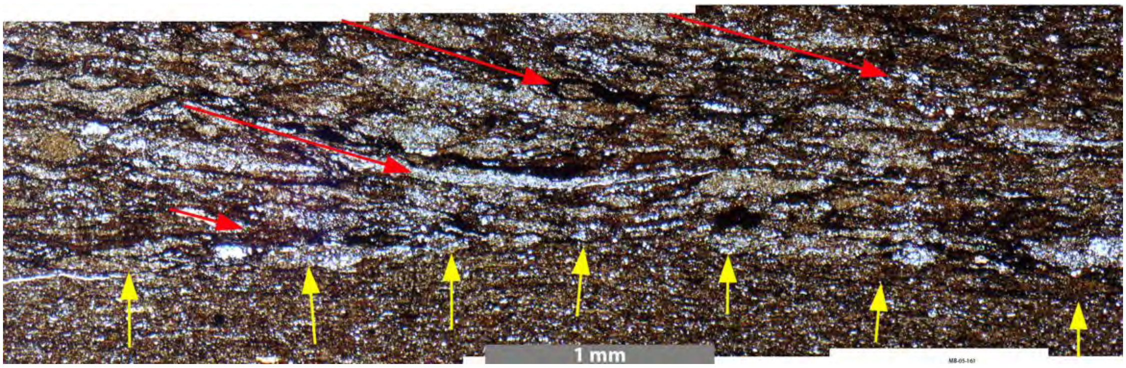

Figure 16. Detail of a photomicrograph mosaic (Marmbulligan-1) that shows a lenticle-rich layer (base marked with yellow arrows) with an inclined (red arrows) internal fabric. An interpretation of this feature as a reflection of traction transport by bottom currents is supported by experimental work and studies of other shale successions (e.g., Schieber et al., 2019; see discussion below).

CS3: Striped Calcareous Carbonaceous Shale and CS4: Faintly Striped Calcareous Carbonaceous Shale

The striped calcareous carbonaceous shale (CS3) and faintly striped calcareous carbonaceous shale (CS4) are in their physical aspects (striped appearance, clay drapes, carbonaceous laminae of the parallel laminated, wavy-anastomosing, and homogenous-lenticular variety) quite comparable to facies CS1 and CS2, and are indistinguishable on visual inspection alone (Online Supplement 6, available as AAPG Datashare 216 at www.aapg.org/datashare). The main difference is that they contain enough early diagenetic carbonate (dolomite, ankerite) to noticeably enhance rock hardness. When scratched with a hardened steel scribe they

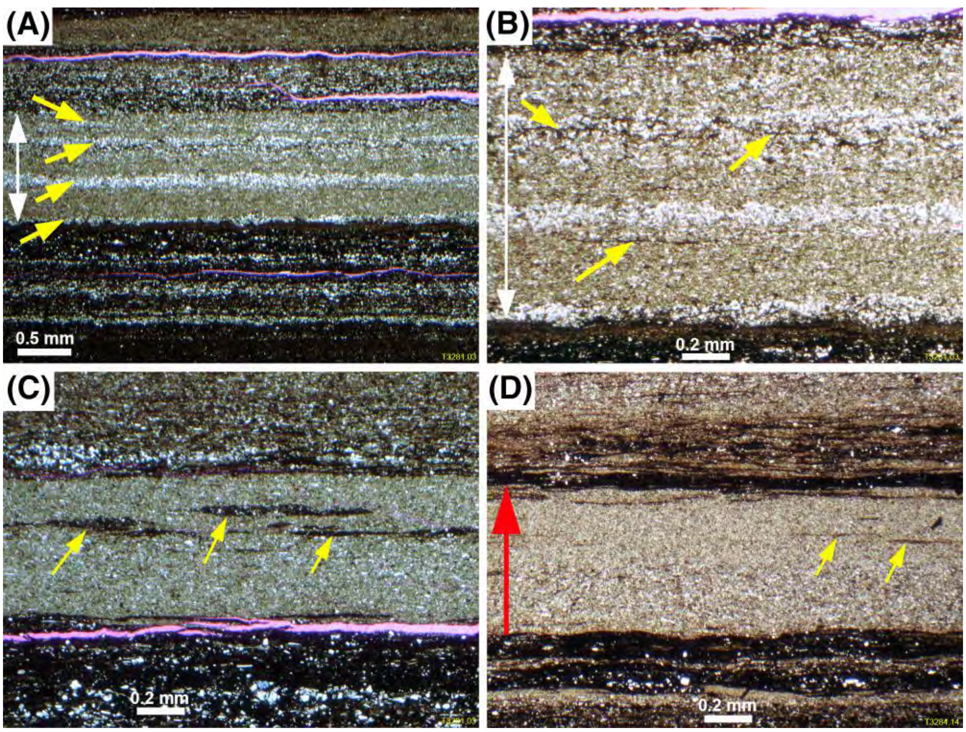

Figure 17. Photomicrographs showing sedimentary features of clay drapes within facies CS1 (striped carbonaceous shale) (Tanumbirini-1). (A) Clay drape with thin silt laminae (yellow arrows). (B) Close-up view of (A). Silt laminae vary in thickness laterally and consist of small adjoining lenses. Thin carbonaceous streaks (yellow arrows) are common. (C) Clay drape with well-developed dark carbonaceous streaks (yellow arrows). Internally these streaks are textually identical to wavy-anastomosing carbonaceous laminae (Figure 12). (D) Clay drape with subtle vertical grading (red arrow), due to a higher content of silt grains in the lower half. It also contains dark carbonaceous streaks (yellow arrows) as seen in (C).

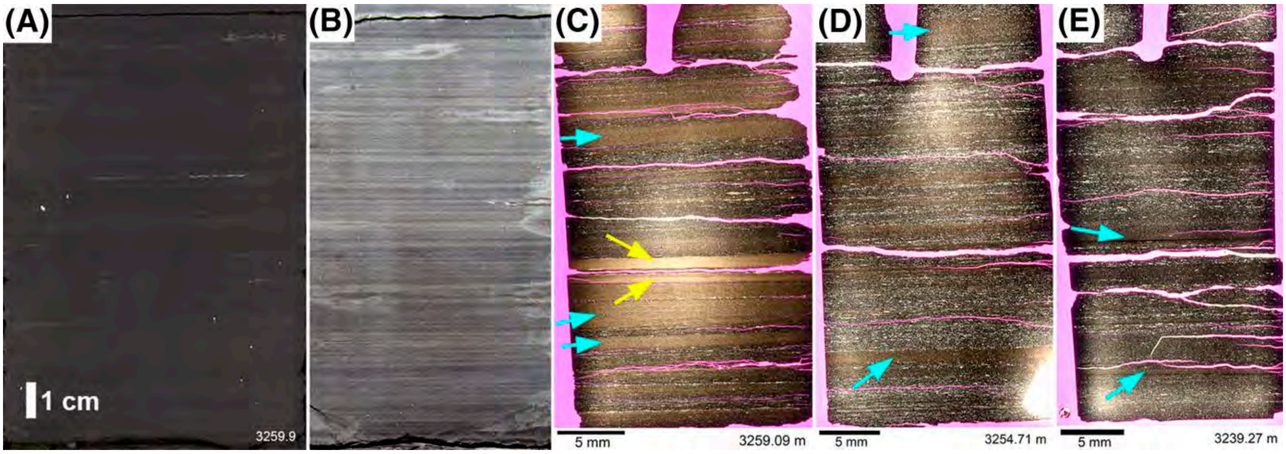

Figure 18. (A) Original core photograph of typical facies CS2 (faintly striped carbonaceous shale) (Tanumbirini-1) that reflects what was visible at the start of core examination. (B) A strongly contrast-enhanced image that was taken after the core had been carefully washed. Lamination is now clearly visible, though still not as pronounced as lamination in CS1 (striped carbonaceous shale) (Figure 10). (C) A thin section from that core interval, with two clear clay drapes (yellow arrows). (D, E) Thin sections from the highest TOC (total organic carbon 5–8 wt. %) part of the core. Potential clay drapes are thin and inconspicuous. Blue arrows mark layers that are "nonblack" but are not clay drapes.

cannot be grooved as deeply as facies CS1 and CS2, a difference that is corroborated by increased Leeb rebound hardness. Even though CS1 and CS2 contain scattered carbonate grains, they are low in abundance (<0.1 vol. %) and when HCl is applied, there is no visible reaction (gas bubble production). In CS3 and CS4, however, bubbles are readily observed at carbonate content >10 vol. % and still observable at carbonate contents of a few percent. Figure 19 illustrates the visual appearance of facies CS3 and CS4. The carbonate component that sets apart facies CS1 and CS2 from calcareous equivalents CS3 and CS4, is best documented in argon ion-milled SEM samples (Figure 20), because many of the carbonate grains are smaller than the nominal thickness of the thin sections and thus either missed or under-reported when viewed by petrographic microscope.

Soft-Sediment Deformation in Striped Shales

Soft sediment deformation tends to be rather subtle, manifesting as deformed layering, dip differences between superjacent shale intervals that may indicate slump related shear planes (Online Supplement 4B, available as AAPG Datashare 216 at www.aapg.org/datashare), and as truncation surfaces in association with mineralized fractures that suggest deformation and shearing of semiconsolidated sediment (Online Supplement 4C, available as AAPG Datashare 216 at www.aapg.org/datashare).

In addition, pinch and swell features of bundles of laminae (Online Supplement 4D, available as AAPG Datashare 216 at www.aapg.org/datashare) occur in places and resemble what has been described in the literature as "loop structures" (e.g., Bradley, 1930, 1931; Cole and Picard, 1974). Boudinage-like features of this type suggest intrastratal ductile deformation (stretching) in a semicohesive sediment (Calvo et al., 1998).

Aforementioned soft-sediment deformation features were largely observed in Tanumbirini-1, a core drilled comparatively close to the inferred basin margin (Figure 2). This core was exceptionally clean and complete when viewed in 2016, but when Juergen Schieber reexamined the core at the NTGS core facility in 2022, it showed considerably degradation due to pyrite oxidation in a warm and humid climate, making observation of such subtle details considerably more difficult. On the whole, the reported features probably represent less than 0.1% of the striped shales (Figure 6) in Tanumbirini-1. In core images and cores of comparatively more distal wells (Tarlee-1, Tarlee-S3, Lady Penrhyn-1, McManus-1), the above features were not detected.

Truncations and Scours in Striped Shales

When closely examined, the striped shales show erosion surfaces in multiple places where laminae are truncated and then draped over by the same type of

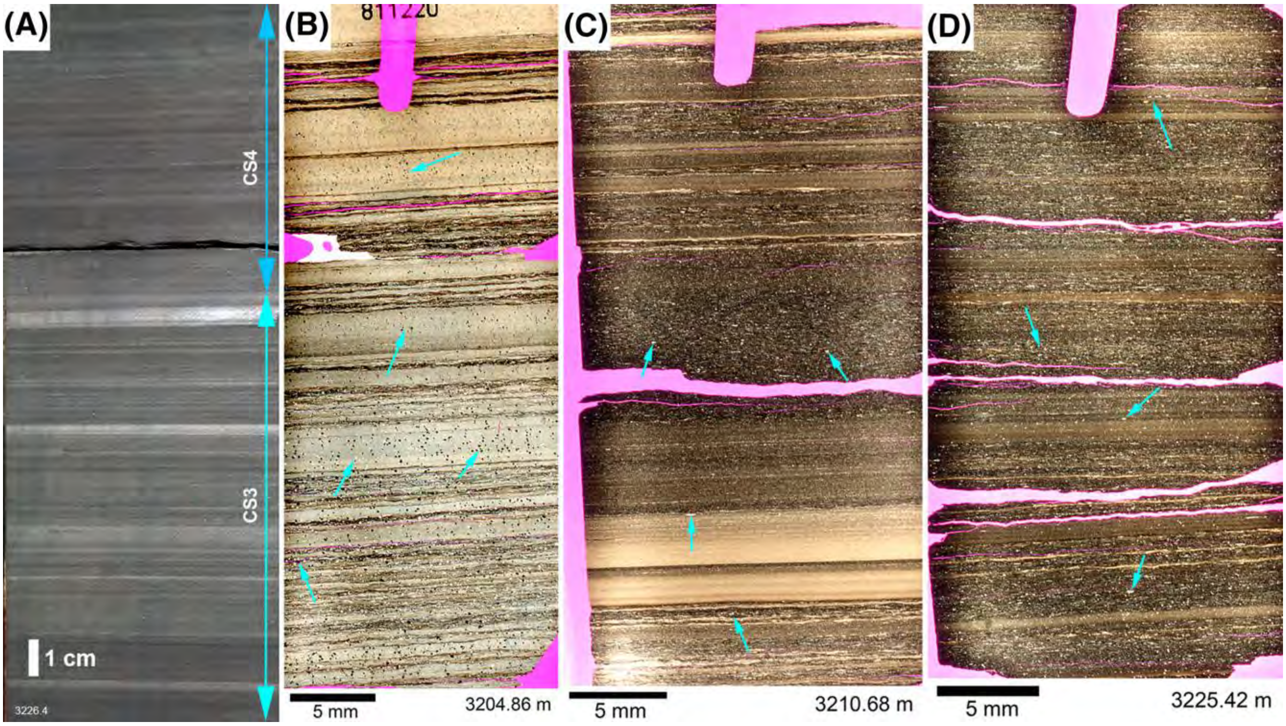

Figure 19. (A) Core image (washed, contrast enhanced) that illustrates the difference in appearance between CS3 (striped calcareous carbonaceous shale) (visible light colored clay drapes) and CS4 (faintly striped calcareous carbonaceous shale) (largely devoid of clay drapes). (B) Example of CS3 with abundant clay drapes, and abundant dark spots of pyrite and light specks of carbonate (blue arrows) that are disseminated through the rock. (C) Example of CS3 with comparatively low carbonate content (scattered tiny light specks of carbonate, blue arrows) and clay drapes. (D) Example of CS4 devoid of clay drapes. Thin "nonblack layers" (comparable to those seen in facies CS2 [faintly striped carbonaceous shale] Figure 18D, E) consist of a mixture of rip-up clasts (lenticles), microbial mat fragments, and variable amounts of silt. Images (B), (C), and (D) are photomicrographs.

shale that underlies the truncation surface. Examples are shown in Figure 21.

Cemented Beds

These types of beds occur interspersed with the dominant striped shales and two facies are distinguished, dark mudstone (DM) and cemented mudstone (CM) (Figure 5). Facies DM typically occurs as centimeter to decimeter thick beds at a spacing of decimeters to meters, whereas facies CM is comparatively rare and was only observed in the 3260 to 3270 m interval of the Tanumbirini-1 core (see detailed stratigraphic section made available as Online Supplement 1, available as AAPG Datashare 216 at www.aapg.org/datashare).

DM: Dark Mudstone Facies

Facies DM contrasts with associated lithologies (CS1, CS2, CS3, CS4) by forming layers that are distinctly thicker (5 mm to as much as 0.2 m) and lacking millimeter-scale layering (Figure 22A). Even in thin section, facies DM is rather homogeneous (Figure 22B), though faint vestiges of primary lamination may be preserved in places (Figure 22C). Homogenous appearance and a general lack of sedimentary structures renders facies DM beds hard to interpret at the core description level. Petrography, specifically SEM examination of ion milled surfaces, is needed to get a deeper understanding of these rocks.

Under the SEM, DM samples show randomly scattered ankerite grains and grain clusters (Figure 23), as well as a rather faintly developed wispy internal layering marked by horizontal lenses of ankerite. In addition, there are irregular patches and streaks of interstitial OM (black in backscattered electron SEM images). At higher magnification (Figure 23, inset), dolomite and clusters of phosphatic spheres (like in Figure 14D) are visible.

Tiny grains of diagenetic quartz permeate the clay-OM matrix of this rock type (Figure 24), and are by volume the dominant diagenetic cement

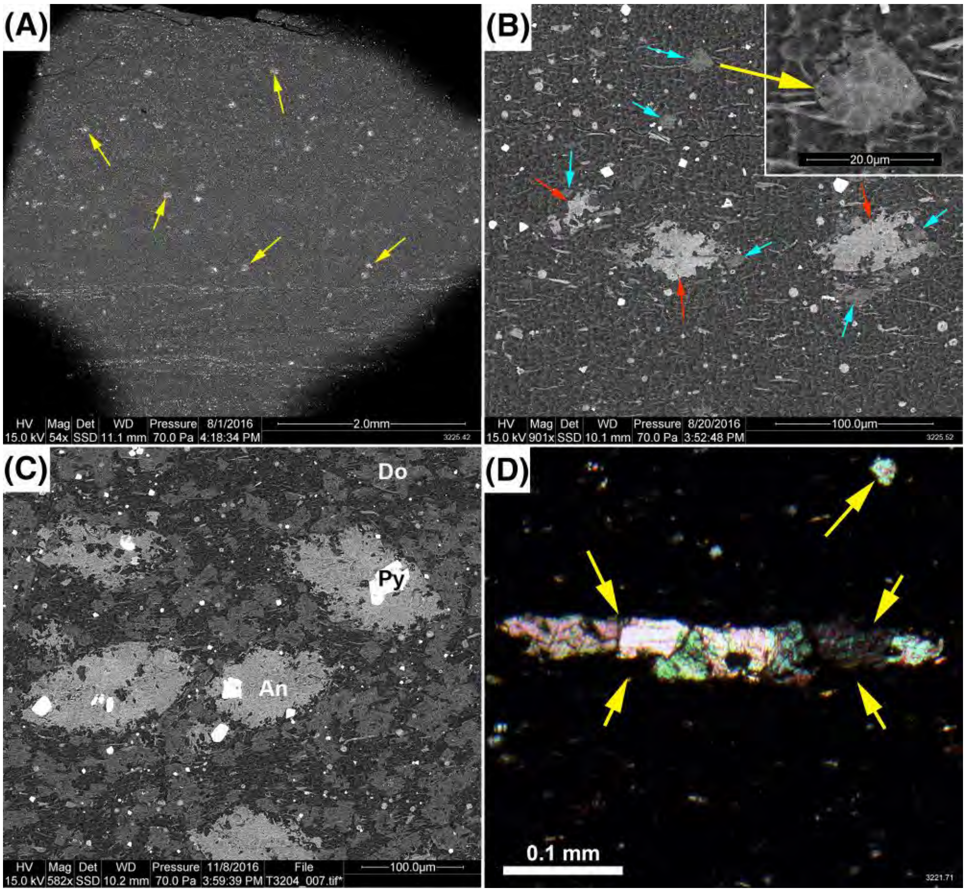

Figure 20. Scanning electron microscope (SEM) (A–C) and petrographic microscope images (D) of carbonates in CS3 (striped calcareous carbonaceous shale) and CS4 (faintly striped calcareous carbonaceous shale). (A) Overview image of CS4 sample with scattered spots of ankerite (yellow arrows). This sample has only a few vol. % carbonate content. (B) A closer view of early diagenetic carbonate grains in the same sample. Two carbonate types are present, dolomite (Do, blue arrows) and ankerite (An, red arrows). Dolomite grains (see upper right inset) tend to be euhedral and largely free of inclusions, whereas ankerite grew more irregularly and has inclusions of detrital particles and dolomite. (C) For comparison, a view of the matrix of a CS3 sample with high carbonate content. Light gray ankerite grains contain inclusions of pyrite (Py), quartz, and (in places) dolomite. Dolomite has smaller and often euhedral grains that are disseminated through the matrix and in places shows inclusions of pyrite. (D) A lens of dolomite (yellow arrows) viewed with crossed polarizers under the petrographic microscope.

(Figure 24A, B). Petrographic study suggests a diagenetic to detrital quartz ratio of 10 or higher (Figure 24A). Larger diagenetic quartz grains (micron-size) tend to be more euhedral (Figure 24C), whereas smaller ones (tens to hundreds of nanometers) generally have rounded-irregular outlines (Figure 24D).

Pyrite Concretions Associated with DM Beds—Pyrite layers and lenses are common in certain sections of core and range in thickness from a few millimeters to as much as much as 30 mm (Online Supplement 7, available as AAPG Datashare 216 at www.aapg.org/datashare). They typically occur within homogenous beds (DM), and are interpreted to record a hiatus or slow-down of sedimentation (marking the tops of parasequences, Lazar et al., 2015b). Because pyrite nodules and layers need to grow over considerable time spans to reach a size of several centimeters (Canfield and Raiswell, 1991; Schieber and Riciputi, 2005; Schieber, 2011), their association with DM beds is consistent with the sediment-starved condensed section scenario envisioned for the formation of DM intervals.

CM: Light Colored Cemented Mudstone with Ankerite, Dolomite, and Silica Cements

Although diagenetically cemented beds of the DM type are common throughout the Tanumbirini-1 and Marmbulligan-1 cores, beds of facies CM are less common and even more intensely cemented. Common cementing minerals are ankerite and dolomite, as well as disseminated diagenetic quartz. These beds differ from DM in that they are harder and of lighter color (Figure 25), owing to a greater abundance of light-colored cement minerals (dolomite, ankerite, quartz; Figure 26A, B) and lower OM (bitumen streaks) content.

Cross-cutting and inclusion relationships identify pyrite as the earliest-formed diagenetic mineral, closely followed by euhedral and inclusion-free dolomite. Pyrite and dolomite occur as inclusions in ankerite with multiple growth generations. Detrital phyllosilicates in the matrix show differential compaction around ankerite clusters, attesting to the precompaction origin of carbonate cements. Ankerite may show inclusions of fine crystalline (nanometer to micron scale) diagenetic quartz. The latter is by volume the dominant cement and differential compaction of the phyllosilicates around these quartz grains suggest an early diagenetic (precompaction) origin as well. Although some diagenetic quartz is overgrowing detrital cores (Figure 26C, D), most of it appears to be a primary precipitate.

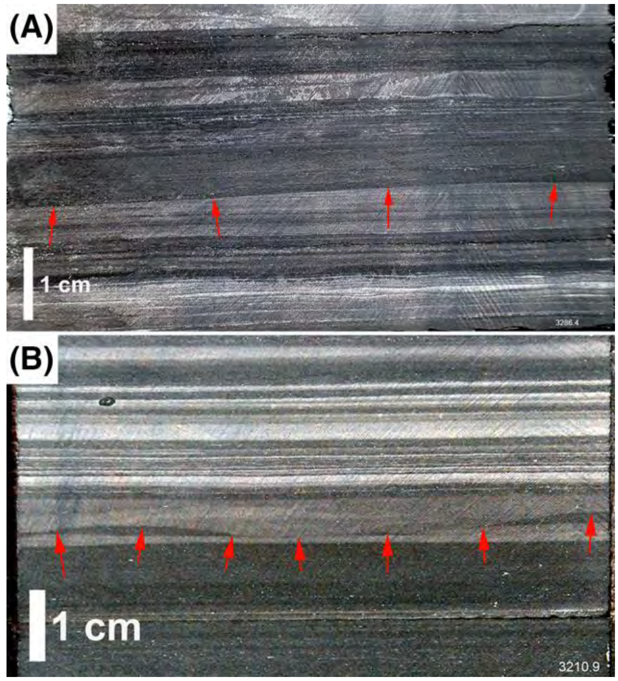

Figure 21. Examples of truncation surfaces. (A) CS1 (striped carbonaceous shale with erosion surface marked by red arrows. (B) Undulose erosion surface (marked by red arrows) in calcareous striped shale (CS3 [calcareous carbonaceous shale]).

OBSERVATIONS FROM THE AMUNGEE MEMBER IN OTHER CORES

The Tanumbirini-1 and Marmbulligan-1 cores are located comparatively close to each other (25 km distance, Figure 1), some distance offshore from a depocenter/delta complex. Other cores drilled at significant distance (Tarlee-1, 190 km; Tarlee-S3, 200 km; McManus-1, 120 km; Lady Penrhyn-1, 150 km; see Figure 1 for locations) from the former two cores still show remarkably similar facies characteristics (Figure 27). Thus, the facies characteristics of the striped shales that constitute the bulk of the black shales in the Amungee Member persist over distances of hundreds of kilometers and may have covered an area in excess of 50,000 km^2 in the distant past.

Although in the case of the comparatively proximal Tanumbirini-1 and Marmbulligan-1 cores a range of extrabasinal silt and mud layers have been identified and related to sediment input from the basin margins (Figure 6), core photographic examination of distal cores, e.g., Tarlee-S3 and Lady Penrhyn-1, do not show the full range of silt layer types shown in Figure 6. In distal cores silt rich intervals are dominated by ST-ML, together with silt layers (ST) and lenses.

PROCESS-FOCUSED INTERPRETATION

Assessment of depositional environments relies on observation of (1) sedimentary structures for identification of basic processes (transport mode, depositional thresholds, energy levels and fluctuations); and (2) the lateral facies context within the basin. In the case of shales, the first element is informed by observations made on thin sections and hand specimens, and the second element relies on lateral correlations using, for example, wire-line logs and core descriptions. For element (1), the following sections focus on first order textural interpretations of the above-described facies types. For element (2), the implications of these interpretations are considered from a basin-scale perspective.

First Order Interpretation of Striped Shales

Physical sedimentary attributes, such as wavy-anastomosing carbonaceous laminae (Figure 12), lenticular textures (Figure 13), and parallel silty laminae

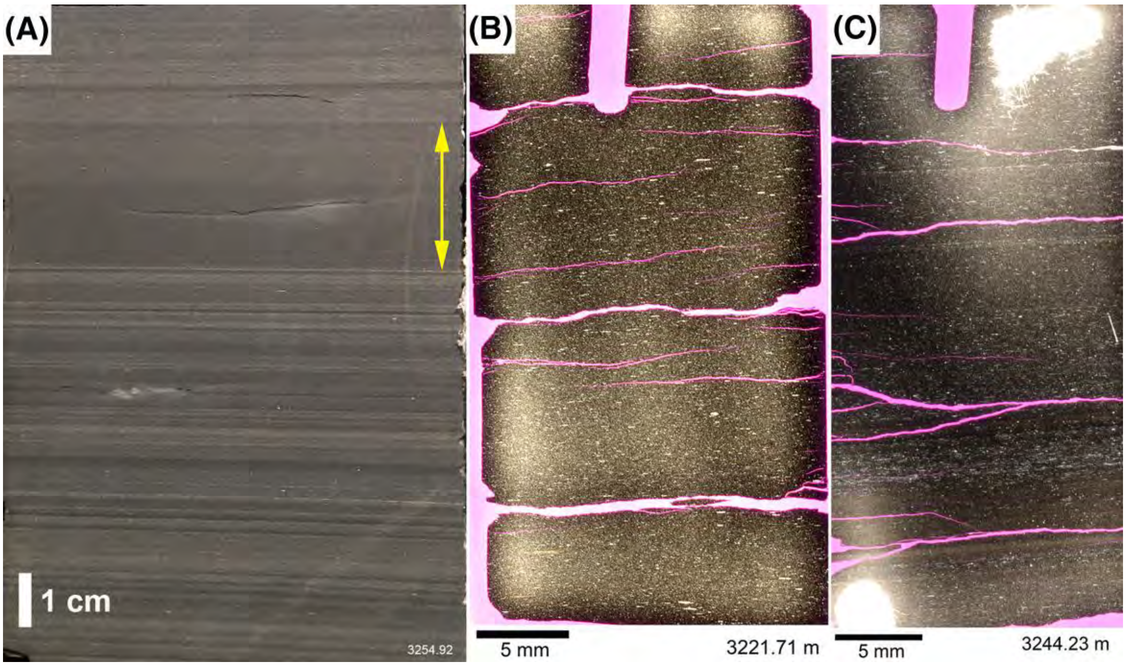

Figure 22. Facies DM (dark mudstone) in core and thin section. (A) Core specimen shows the contrast between homogenous DM layer (yellow double arrow) and associated CS3 (striped calcareous carbonaceous shale) lithology. (B) Homogenous DM in thin section view; light colored lenses are mostly composed of ankerite, similar to those in Figure 20D. (C) Mostly homogenous DM in thin section view; subtle layering is due to compositional differences of original layers.

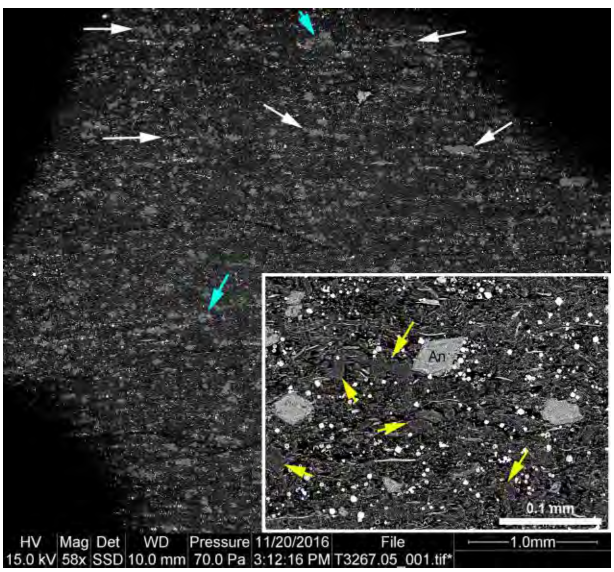

Figure 23. SEM (scanning electron microscope) view (backscatter mode) of DM (dark mudstone) sample that shows rather faintly developed wispy internal layering. White arrows point to horizontal streaks of ankerite, blue arrows point to clusters of ankerite grains. Inset shows quasi euhedral ankerite grains (An) and dolomite grains (yellow arrows) scattered through the matrix, as well as scattered, bright, micron-size phosphate spheres.

(Figure 10) are shared by all striped shale subtypes. Wavy-anastomosing carbonaceous laminae from multiple Mesoproterozoic shale successions that essentially are identical to those seen in the Amungee Member were previously interpreted as the remains of benthic microbial mats (e.g., Schieber, 1986, 1989, 1999b, 2007a, b; Patranabis-Deb et al., 2007). These literature examples are indistinguishable from images of the Amungee Member striped shales, including minute textural attributes like the tiny phosphate spheres (Figure 14D), considered to be mineralized bacteria that reflect microbial decay of the original mats (Schieber et al., 2007b).

Photosynthetic microbial mats on muddy substrates have one particular "weakness"—their substrate is water-rich and mechanically weak, causing mats to detach and float up because of gas bubble development (Figure 28). On occasion, this propensity of microbial mats has been called upon to cast doubt (e.g., B. Schopf, 1982, personal communication) on the model proposed by Schieber (1986), used here to explain the origin of wavy-anastomosing

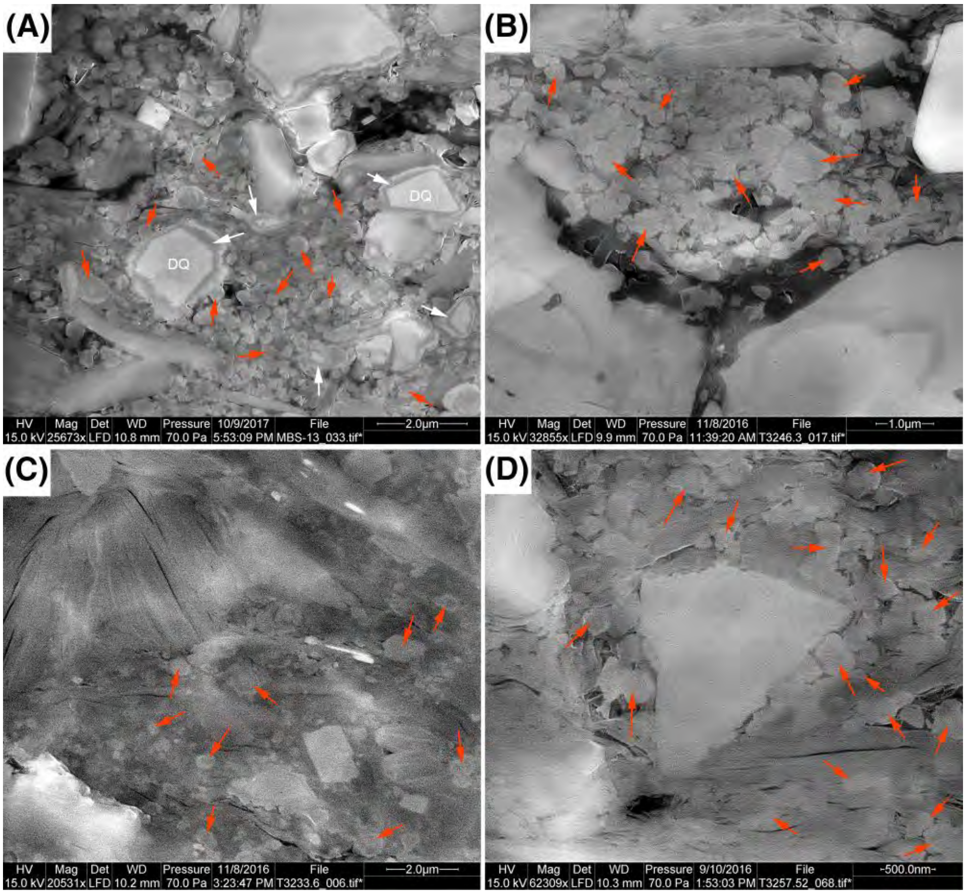

Figure 24. SEM (scanning electron microscope) images of fine crystalline quartz (secondary electron mode) in the matrix of facies DM (dark mudstone) (A, B) and facies CS4 (faintly striped calcareous carbonaceous shale) (C, D). (A) Very fine detrital quartz grains (DQ) with overgrowth rims (white arrows) and abundant submicron early diagenetic quartz scattered through the clay matrix (red arrows). (B) Clay-rich matrix with abundant submicron early diagenetic quartz (red arrows). (C, D) For comparison, the clay rich matrix in facies CS4 also contains disseminated submicron early diagenetic quartz (red arrows), but at notably lower abundance than in facies DM.

carbonaceous laminae (Figure 12) and carbonaceous streaks within clay drapes (Figure 17) and reworked beds (Figures 13, 15).

The viability of photosynthetic microbial mats on muddy substrates has been examined experimentally (Schieber, 2021). A mat was grown in a flume channel with intermittent current flow and pulses of silt and clay that covered the mat surface. After each sediment pulse the mat re-established itself on the new sediment surface. At light fluxes of ≈200 lux (lx) (adequate for reading) the mat stayed attached to the substrate, only developing small pinnacles in areas of gas accumulation (decay gases) from the underlying sediment. When light flux increased to ≈650 lx, photosynthesis ramped up, and abundant gas bubbles in the surface mat floated up segments of the mat and disrupted the previously continuous mat cover. At light fluxes of 700 to 800 lx the mat became highly productive and deep green, and even though it domed and floated intermittently, it still formed a covering layer. Diagnostic sedimentary structures (mat roll-ups, flip-overs, resedimented microbial mat fragments; Schieber, 1986) were recorded and provide a close analogue to ancient counterparts (Figure 29). Unlike the flume studies that explored parallel lamination and lenticular fabric (Schieber et al., 2007a, 2010; Yawar and Schieber, 2017), the flume experiments on benthic mat growth have not yet been formally published (Schieber, 2021), but key aspects are summarized in Figure 29. In modern oceans, the lighting conditions for a thriving but nondetaching photosynthetic mat should easily be met in the 50 to 200 m water depth range. For Proterozoic epicontinental basins it is therefore very feasible that photosynthetic microbial mats covered large parts of muddy

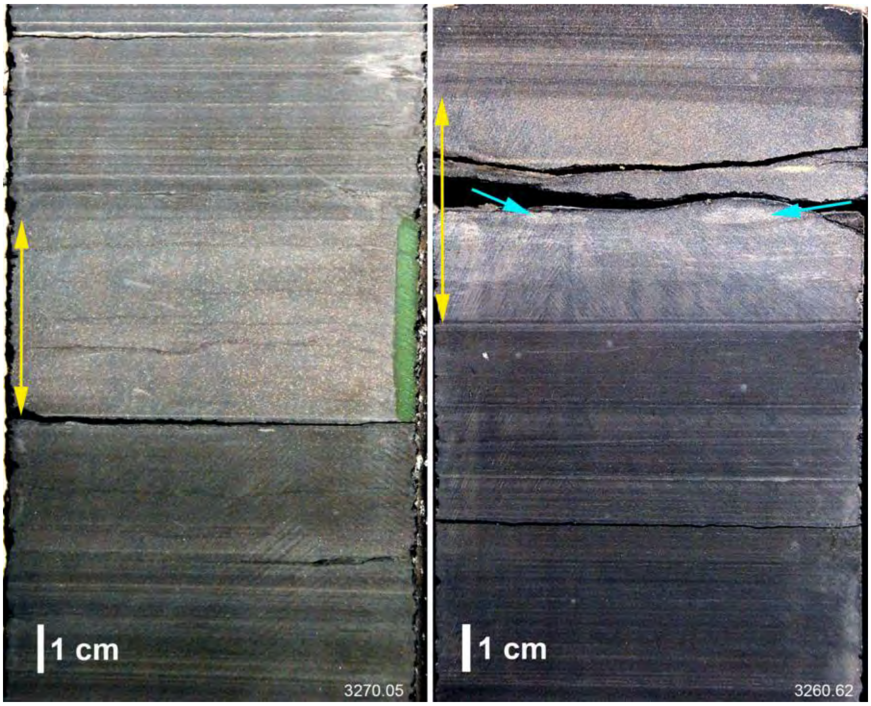

Figure 25. Two drill core examples of light-colored cemented beds of facies CM (cemented mudstone (marked by yellow double arrow). Although the general texture is homogenous and lacks lamination, blue arrows point to more intensely cemented parts, possible areas of initial concretion growth.

seafloors and generated primary OM for direct incorporation into the accumulating sediment.

In previously published examples of striped shales (e.g., Schieber, 1986, 1989; Sur et al., 2006; Patranabis-Deb et al., 2007), carbonaceous layers of microbial mat origin (wavy-anastomosing carbonaceous laminae) strongly dominated the shales in question, whereas in the Amungee Member carbonaceous layers are dominated by the lenticles illustrated in Figures 13 and 14. Textural equivalents occur in other Mesoproterozoic shale units (e.g., Sur et al., 2006; Patranabis-Deb et al., 2007; Schieber, 2007a) as well as in younger shale successions (e.g., Li and Schieber, 2018; Li et al., 2021). Experimental studies (Schieber et al., 2010; Schieber, 2011) suggest that the lenticles resulted from bottom currents that eroded/cannibalized penecontemporaneous muddy substrates in the Beetaloo Sub-basin and generated sand-sized soft rip-ups that were redistributed by currents across the basin floor. The length-to-width ratio of lenticles is consistent with them being derived from muddy substrates with water contents of approximately 80 vol. %. Inclined fabrics, as seen in Figure 16, suggest bedload transport (in the form of ripples) for lenticle-rich layers. Comparable features were observed in other case studies (Schieber et al., 2019). A part of the eroded material likely disintegrated initially, followed by flocculation and redeposition in the course of floccule ripple migration (Schieber et al., 2007a; Schieber, 2011). The parallel coarse silt laminae shown in Figure 14 closely resemble experimentally produced textures that record the simultaneous passage of coarse silt ripples and floccule ripples over the seabed (Yawar and Schieber, 2017), an interpretation consistent with the lens-like attributes (microripples) of these laminae.

The clay drapes suggest episodic introduction of fine-grained sediment from an external source, likely a low relief hinterland that facilitated physical weathering of low grade metamorphic rocks (abundance of detrital chlorite, Figures 9, 26B) to mud grade. Their internal features (grading) suggest episodic incursion of muddy suspensions, either as distal extensions of fine-grained gravity flows (delta-front turbidites, hyperpycnites, wave-enhanced sediment gravity flows), or possibly as nepheloid flows and settling from interflows (Stanley, 1983; Bhattacharya and MacEachern, 2009; Macquaker et al., 2010a). Bottom currents redistributed these materials to more

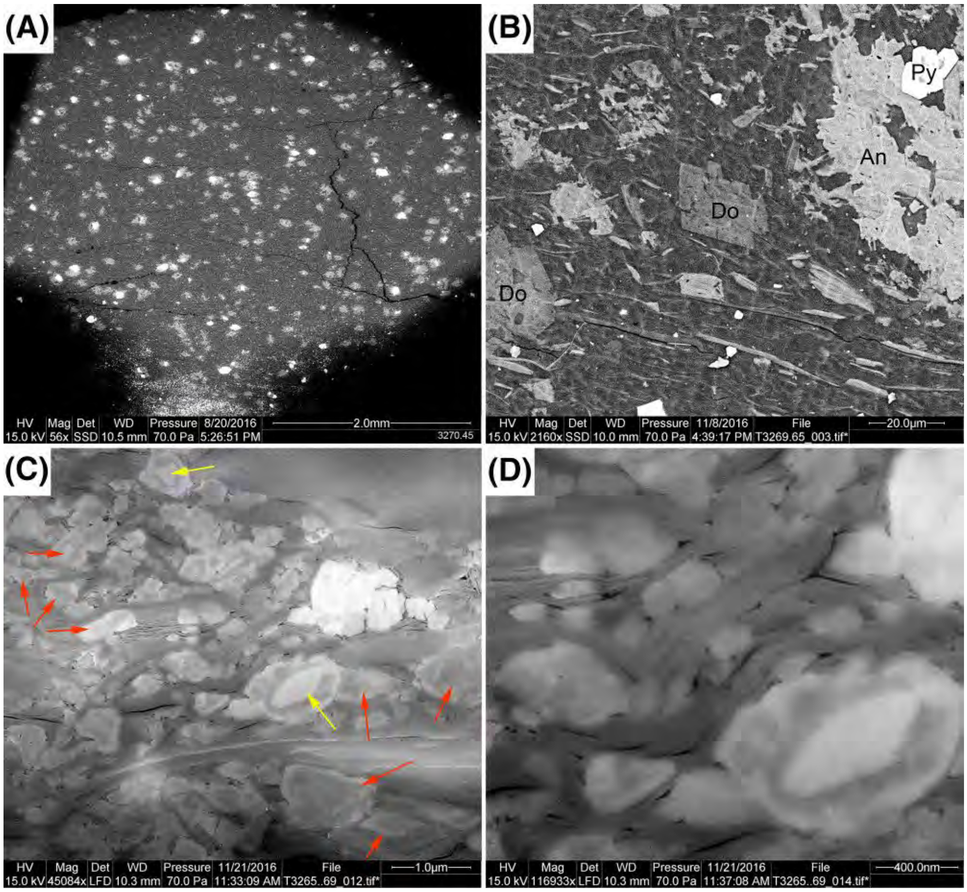

Figure 26. (A) Abundant bright ankerite (An) spots in a CM (cemented mudstone bed SEM [scanning electron microscope] backscatter image). (B) Closer view showing euhedral character of dolomite (Do) relative to the more irregular and inclusion-rich ankerite. Py marks pyrite grain. Note abundant bright chlorite flakes. (C) Abundant fine crystalline diagenetic quartz (red arrows) mingled with clays, as well as detrital quartz with overgrowth rims (yellow arrows). (D) Detail from (C), showing differential compaction and deformation of phyllosilicates (illite, micas) around quartz grains, as well as an inclusion-free overgrowth rim on a detrital quartz grain.

distal reaches of the basin, incorporated eroded microbial mat fragments (dark streaks), and produced parallel silt-mud laminites due to simultaneous bedload transport of coarse silt, flocculated mud, and water-rich fine-grained rip-ups (Yawar and Schieber, 2017).

When compared to CS1, CS2 is a type that likely records a more distal depositional setting, as indicated by very thin to absent clay drapes. Yet, in spite of a more distal setting, CS2 records a comparable bottom current regime, and even records, via particle size variability (Figure 13), different levels in the strength of eroding and transporting currents.

Calcareous striped shales (CS3 and CS4) mainly differ from CS1 and CS2 by their larger content of early diagenetic carbonates. The latter characteristic suggests lower sedimentation rates relative to CS1 and CS2, with early diagenetic cements filling a larger share of the available pore space (e.g., Taylor and Macquaker, 2000; Machent et al., 2007).

The striped shales record an array of depositional processes that were active simultaneously as well as intermittently in a comparatively distal environment. Processes include intermittent event sedimentation (clay drapes), bedload transport of silt and flocculated OM-clay aggregates, seafloor erosion and bedload transport of rip-up clasts, and intermittent growth of microbial mats. Erosion surfaces (Figure 21) attest to bottom currents capable of eroding surficial muds at bed shear stress values between 0.2 and 0.4 Pa (Schieber et al., 2010; Schieber, unpublished experimental data), conditions that would support the transport of sand in bedload (Schieber, 2011). These bottom currents carried flocculated material and reworked aggregates across the basin floor, redistributed them over wide areas, and led to a smooth seafloor

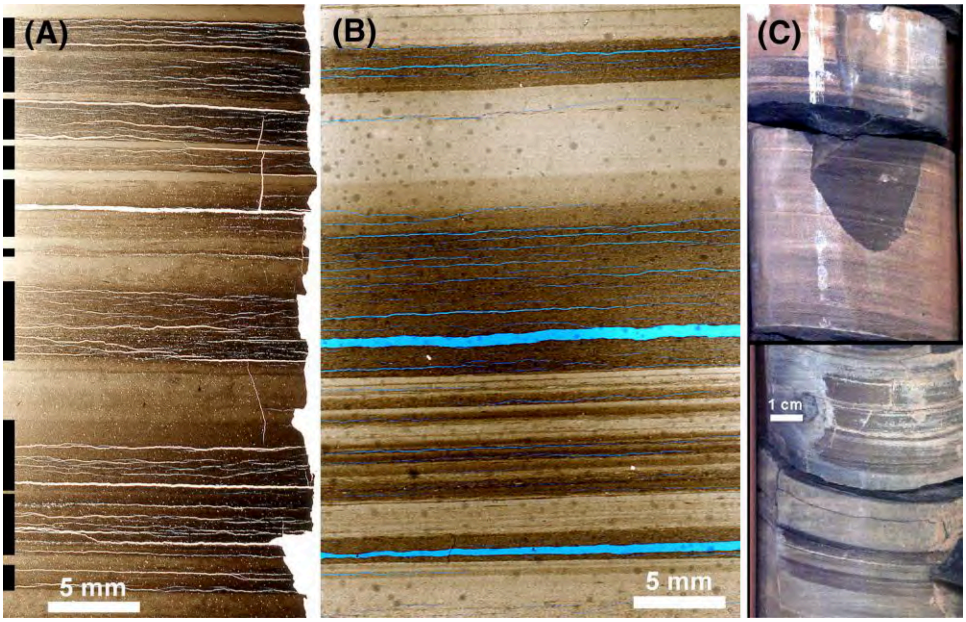

Figure 27. Examples of shale facies in cores drilled at a significant distance from Tanumbirini-1 and Marmbulligan-1 (Figure 1). (A) Thin section scan of striped shale from Tarlee-1. TOC-rich (total organic carbon) layers marked with black bars at left. During core storage the pyrite in carbonaceous layers starts to oxidize to Fe-sulfate, expands, and develops irregular bedding parallel fractures that are filled with Fe-sulfates. (B) Striped shale in thin section from McManus-1 core (round spots are air bubbles in the epoxy that bonds the rock slice to the glass carrier). (C) Pieces of core from Lady Penrhyn-1 that show striped shale characteristics.

topography. Given that the microbial mat aspect of these sediments requires frequent pauses in sediment accumulation, these current systems should probably not be considered shallow water contourites in the classical sense (Viana and Faugères, 1998; Verdicchio and Trincardi, 2008). Given the presumed warm climate in the Mesoproterozoic (de Kock et al., 2024) thermohaline circulation is a potential driving force for bottom water circulation, or alternatively a combination of wind driven and thermohaline circulation (DeReuil and Birgenheier, 2019).

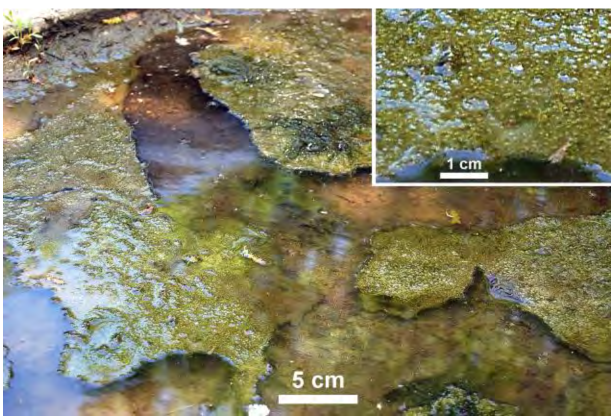

Figure 28. A shallow puddle (approximately 5–10 cm deep) with fragmented microbial mat that has floated to the surface because of gas bubble development. See upper right inset for details.

First Order Interpretation of Cemented Beds

In shales, compaction is one of the main causes for the development of planar fabric and the appearance of layering/lamination at the hand specimen and core scale (e.g., Potter et al., 1980). The high level of early diagenetic cementation observed in DM and CM beds must have substantially reduced postdepositional compaction of these rocks, and is the likely cause for their homogenous appearance and almost complete lack of visible layering (e.g., Otharán et al., 2022). One textural element of cemented beds, dispersed micron-size phosphate spheres, is also a key characteristic of carbonaceous beds in striped shales (Figure 14D) and is not observed in clay and silt beds. This suggests that prior to cementation, the original sediment was strongly dominated by carbonaceous layers and that episodic introduction of fine-grained sediment from external sources was rare or did not occur. As such, the cemented beds can be considered to be starved of terrigenous clastic input

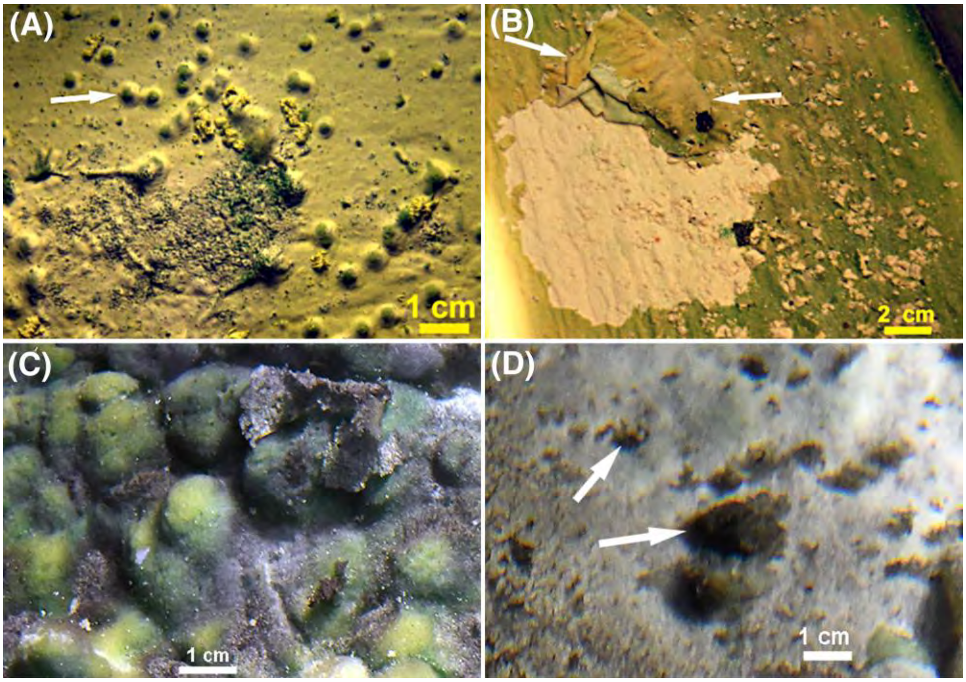

Figure 29. Observations from microbial mat grown in flume channel. (A) Thin mat with trapped gas bubbles (arrow), grown at comparatively low light levels. (B) Current erosion of thin mat. The image shows folded over and rolled up mat between white arrows. The bright spots on the mat surface are holes produced by bursting bubbles, exposing the underlying white clay bed. (C) Thicker mat grown at higher light levels, but still covering the clay bed and not floating up. Although there is substantial surface relief, most of the mat consists of very water-rich extracellular polymer substances. When buried and compacted these layers would be reduced to the wavy carbonaceous laminae seen for example in Figure 12. (D) Fragments of eroded mat (arrows) on top of an exposed clay bed. When buried and compacted, these would become carbonaceous streaks like seen in Figure 17.

(Taylor and Macquaker, 2000, 2014; Macquaker et al., 2007) and that they mark prolonged pauses in sedimentation (Schieber et al., 2017, 2022). The CM beds contain the same cements as beds of DM, but appear lighter because of higher levels of cementation and a corresponding lower content of interstitial OM (bitumen). The carbonate cements (ankerite, dolomite) occur in a phyllosilicate matrix with abundant diagenetic silica, and likely record time periods of extremely slow sediment accumulation (condensed record; e.g., Milliken and Day-Stirrat, 2013; Lazar et al., 2015b).

From a sequence stratigraphic perspective, these highly cemented beds may represent the cemented zones below marine flooding surfaces and help to define parasequences (Taylor and Macquaker, 2000, 2014; Macquaker et al., 2007; Schieber et al., 2016). If we consider the much more common DM beds as possible parasequence tops, then the much more strongly cemented CM beds likely mark transgressive and maximum flooding surfaces.

First Order Interpretation of Extrabasinal Silts and Muds

The GM beds and SMC contrast sharply with their host sediment, the much more thinly bedded/laminated striped shales, suggesting that they record rare events rather than everyday sedimentation. Grading, basal cross-laminated silt, rip-up particles derived from the underlying carbonaceous shales (carbonaceous streaks, Figure 8C), soft-sediment deformation (Online Supplement 2, available as AAPG Datashare 216 at www.aapg.org/datashare) and convolute bedding (Online Supplement 4, available as AAPG Datashare 216 at www.aapg.org/datashare), suggests rapid deposition from fast moving turbulent suspensions. The vertical succession in SMC beds, from bedload-deposited ripples to suspension-settled muds can be interpreted as base-truncated (C-D-E) Bouma sequences, and GM beds as a more distal extension once all silt has been deposited (e.g., Reineck and Singh, 1980; Allen, 1985). The observation of soft-sediment deformation and slump-like features in thick GM and SMC beds (see Online Supplement 1, available as AAPG Datashare 216 at www.aapg.org/datashare) is consistent with such an interpretation. Graded SMC could also be interpreted as storm layers (e.g., Aigner and Reineck, 1982; Schieber, 1986, 1989), but the thicker layers (>1 cm) are difficult to explain with such a model. The GM and SMC beds have rather uniform composition (Figures 8D, E; 9), suggesting well-mixed terrigenous input, most plausibly fluvial. Delta-front low density turbidites or hyperpycnites (Bhattacharya and MacEachern, 2009) are likely scenarios. Loop structures in association with other soft-sediment deformation features (Online Supplement 4, available as AAPG Datashare 216 at www.aapg.org/datashare) are suggestive of downslope forces within the accumulating sediments, favoring a delta-front turbidite origin for the GM and SCM beds.

Other silt-rich layers, such as ST, ST-ML, and ST-SM layers (Online Supplements 2, 3, available as AAPG Datashare 216 at www.aapg.org/datashare), do not fit a turbidity current model. On the basis of experimental work with clay-silt mixtures (Yawar and Schieber, 2017), the planar lamination, lens formation, lateral pinch and swell of laminae, and thin interstratified mud drapes of ST, ST-ML, and ST-SM layers suggests simultaneous bedload transport via ripple migration of coarse silt and flocculated mud, probably driven by bottom currents. Even though ST, ST-ML, and ST-SM intervals are linked via their components to sediment sources along the basin margin, bottom currents carried them to their ultimate site of deposition. As such, neither of these silt rich intervals can be used to infer proximality.

Likewise, bundles of parallel silt laminae (Online Supplement 2C, E, available as AAPG Datashare 216 at www.aapg.org/datashare) strongly resemble silt laminated muds elsewhere in the rock record (Schieber, 2016b) that record simultaneous bedload transport and lateral accretion of flocculated mud and silt by bottom currents, a transport model that has been validated by flume studies (Yawar and Schieber, 2017).

Regardless of actual thickness, both ST-ML and ST-SM layers consist of a large proportion of submillimeter- to millimeter-thick silt laminae with intervening muddy layers that may be submillimeter drapes to as much as 1 cm thick (Online Supplements 2, 3, available as AAPG Datashare 216 at www.aapg.org/datashare). Cross-laminated silt lenses and ripples (Online Supplement 2, available as AAPG Datashare 216 at www.aapg.org/datashare) signify traction currents and bedload transport. Sharp upper contacts of silt layers and lenses indicate that ST-ML and ST-SM layers were not deposited from decelerating turbulent suspensions (density current). Smooth to undulose bases of silt layers (Online Supplements 2, 3, available as AAPG Datashare 216 at www.aapg.org/datashare) suggest a substrate firm enough to support overriding silt layers. Load casts and ball and pillow structures in ST-SM layers only occur above comparatively thick mud layers, suggesting rapid deposition and high initial water content of the latter (Reineck and Singh, 1980). Thus, ST-ML and ST-SM layers and bundles thereof (Online Supplements 2, 3, available as AAPG Datashare 216 at www.aapg.org/datashare) were deposited rather quickly without significant hiatuses, and may record time periods of sustained bottom current activity, an interpretation supported by extensive flume studies (Yawar and Schieber, 2017; Schieber et al., 2023). Silty sublayers probably reflect bottom current activity at somewhat higher flow velocities (25–40 cm/s at 5 cm flow depth; 0.2–0.4 Pa bed shear stress), whereas intervals with abundant intermingled muddy laminae may indicate somewhat lower flow velocities (15–25 cm/s at 5 cm flow depth; 0.07–0.2 Pa bed shear stress). At times bottom currents were strong enough to erode muddy substrates, as indicated by mud rip-ups observed in some silt beds (Online Supplement 2D, available as AAPG Datashare 216 at www.aapg.org/datashare) and subtle truncation surfaces within striped shale facies (Figure 21). The observation of opposing flow direction in successive rippled silt beds (Online Supplements 2, 3, available as AAPG Datashare 216 at www.aapg.org/datashare) is a good indication that the associated currents were not related to the turbidity currents that brought in the original sediment (GM and SMC) from the basin margins, and supports the operation of bottom current driven (wind driven or thermohaline circulation, see the First Order Interpretation of Striped Shales section) sediment distribution for large parts of the Amungee Member.

At times of increased clastic influx and build-up of a distal turbidite lobes, these bottom currents would have gained access to more silt and undiluted (with OM) muds. As long as the supply lasted, these bottom currents eroded and reworked this resource and carried more silt and mud into distal parts of the basin, and deposited more ST and ST-ML layers than under "normal" circumstances, when bottom currents largely cannibalized basin floor muds and deposited striped shales with lenticular texture. The ST- and ST-ML-enriched intervals in distal shales could be considered as "echoes" of increased clastic supply to the basin margins.

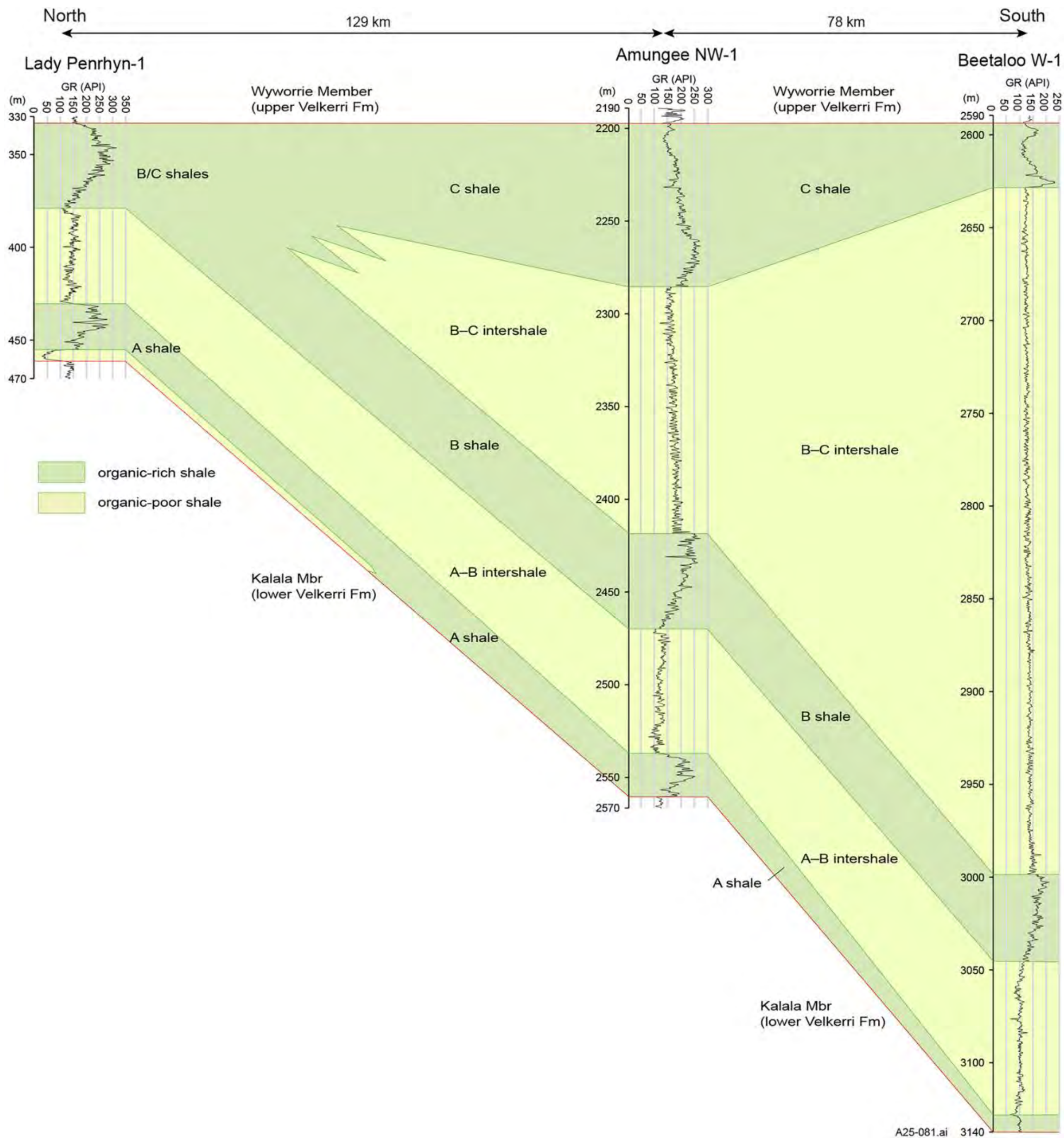

Figure 30. NS profile (profile 1), tracing the TOC-rich (total organic carbon) A, B, and C shales of the Amungee Member, and showcasing a clastic wedge association of the B to C intershale. Gamma log data from well completion reports for Lady Penrhyn-1 (Ledlie, 1989), Amungee NW-1 (Origin Energy Resources, 2016), and Beetaloo W-1 (Origin Energy Resources, 2017), as summarized in Crombez et al. (2023). Fm = Formation; GR = gamma ray.

To summarize, all these silty intervals indicate that bottom currents swept the seabed and redistributed sediment within the basin. Current velocities were variable, with “pure” silt layers indicating somewhat higher flow velocities (allowing for resuspension of mud) and mixed silt-mud laminae indicating smaller flow velocities that allowed simultaneous migration of ripples made from mud floccules and from coarse silt. The source area was dominated by low- to medium-grade metamorphic rocks, and the abundance of detrital chlorite indicates low levels of chemical weathering in the hinterland.

OBSERVATIONS FROM PROFILES 1 AND 2

Using total gamma-ray logs from multiple wells (Figures 1, 2), NS and EW profiles were constructed (Figures 30, 31) to further examine lateral trends in Amungee Member stratigraphy. The TOC-rich intervals (A, B, and C shales) are in most places readily distinguished from the TOC-lean intershale intervals on the basis of gamma ray responses, and we follow in our profiles the gamma-log picks adopted by Crombez et al. (2023) in their basin-wide study of the distribution of these units. The NS profile (profile 1, Figure 30) shows a comparatively uniform thickness for the A and B shales, and a northward thinning of the B to C intershale, illustrating the effect of sediment influx from a southern source on stratigraphic architecture. In addition, Figure 30 shows the distal merger of TOC-rich shales (B/C) in a situation when clastic dilution dwindles to the point of becoming negligible.

The EW profile (profile 2, Figure 31) shows complementary characteristics. The A, B, and C shales

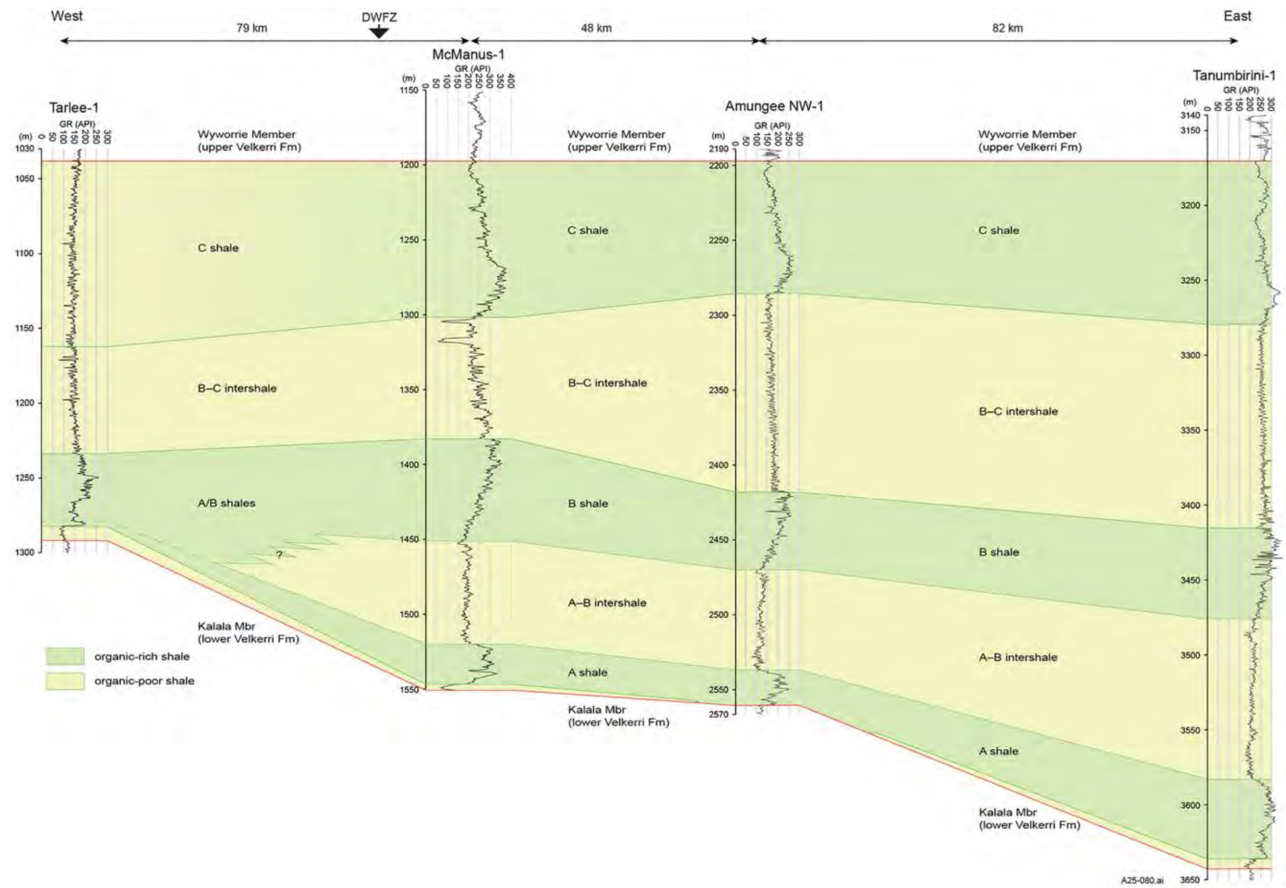

Figure 31. EW profile (profile 2), tracing the TOC-rich (total organic carbon) A, B, and C shales, and pointing to a general westward decrease of clastic supply during the intershale intervals. Gamma log data from well completion reports for Tarlee-1 (Hoffman, 2016), McManus-1 (Lanigan and Ledlie, 1990), Amungee NW-1 (Origin Energy Resources, 2016), and Tanumbirini-1 (Adderley, 2015), as summarized in Crombez et al. (2023). DWFZ = Daly Waters Fault Zone; Fm =Formation; GR = gamma ray.

extend at similar thickness for approximately 200 km; both intershales gradually decrease in thickness westward, and the A and B shales merge at the distal (western) end of the A-B intershale. The differentiation of the C shale in the Tarlee-1 well follows Hoffman (2016). The C shale becomes increasingly organically lean toward the western margin of the subbasin, to the point where it is difficult to distinguish it from the B to C intershale in Tarlee-1. Thus other solutions with regard to correlation appear possible.

DISCUSSION

Although there is a widespread perception that black shales, such as those in the Amungee Member, reflect quiescent conditions in comparatively deep and stagnant basins, recent studies of multiple Phanerozoic examples suggest that this is an overly simplistic perspective (e.g., Schieber, 1998, 2009; Bohacs et al., 2005; Macquaker et al., 2010b; Aplin and Macquaker, 2011; Wilson and Schieber, 2017; Smith et al., 2019). Multiple factors contained in the production—preservation—destruction—dilution parameter space have a bearing on the generation of TOC-rich sediments, and can result in a source rock when these parameters are configured "just right" (Bohacs et al., 2005). In the case of the Amungee Member, sedimentary features such as scours, parallel lamination, rip-up generation, and inclined fabric elements, all of them suggesting pervasive bottom current activity (Schieber et al., 2007a, 2010; Yawar and Schieber, 2017). There likely were interludes of lesser current activity that allowed the intermittent establishment of benthic microbial mats that generated the bulk of the OM that is preserved in these shales. Petroleum geochemistry studies (e.g., Revie, 2017; Jarrett et al., 2019) indicate that the Amungee Member carbonaceous shales contain a mixture of type I and type II kerogens, both of which can be derived from microbial mats (Wang et al., 2023). Given the presumed epicontinental depositional setting, the water plausibly was shallow enough to allow proliferation of photosynthetic benthic microbial mats (Schieber, 1986). Even if the seabed was below photic depth, chemosynthetic mats could still have prospered and aided the accumulation of OM (e.g., Grant and Bathmann, 1987; Jannasch, 1989; Bernard and Fenchel, 1995; Nishino et al., 1998; Schieber, 2007a). Sediment delivery mechanisms within the subbasin were clearly capable of keeping up with accelerated subsidence, and the synoptic, subbasin floor relief was rather flat and likely did not form deep troughs (Schieber, 2016b; Smith et al., 2019). The key elements of the Amungee Member deposystem are summarized in Figure 32.

The Figure 32 schematic implies shoreline attached dispersal systems (deltas in a general sense and associated alongshore clastic deposits), construction of basinward-thinning clastic wedges, and a basin interior mode of sediment dispersion that is largely decoupled from basin margin sedimentation. The toes of clastic wedges are reworked by bottom currents (wind-driven or thermohaline circulation) and dispersed for hundreds of kilometers across the basin floor (lenticular fabric, silt laminated shales), intermingled with microbial mat debris, and interbedded with in situ microbial mat deposits. In the course of ever-alternating erosion, transport, and deposition, muds are spread evenly across the basin floor and lead to rather similar thicknesses of black shale intervals over large distances across the basin interior. The across-basin section lines (Figures 30, 31), support that perspective in as they show clastic dominated sediment wedges (silt and gray shales) that contrast

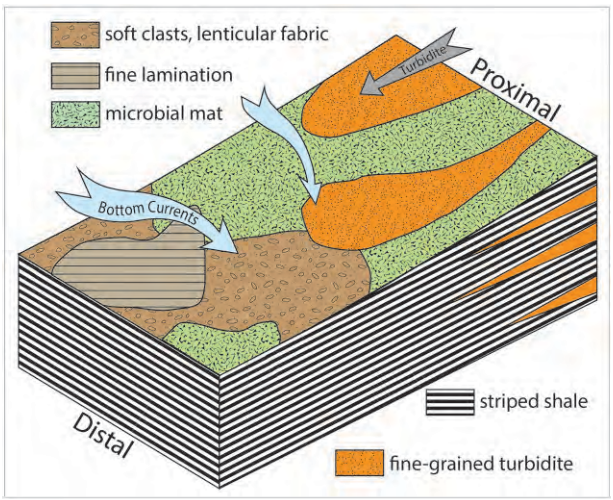

Figure 32. Schematic view of the various processes that operated on the seafloor at the time of deposition of the Amungee Member. Benthic microbial mats were primary producers of organic matter. Bottom currents eroded mats and other muddy substrates, as well as producing parallel lamination and lenticular fabric. Near the basin margin, the distal parts of fine-grained turbidites and hyperpycnites are intercalated with basin floor mudstones. Via multiple interludes of reworking and remixing, they provided the clastic component found in the basin floor muds.

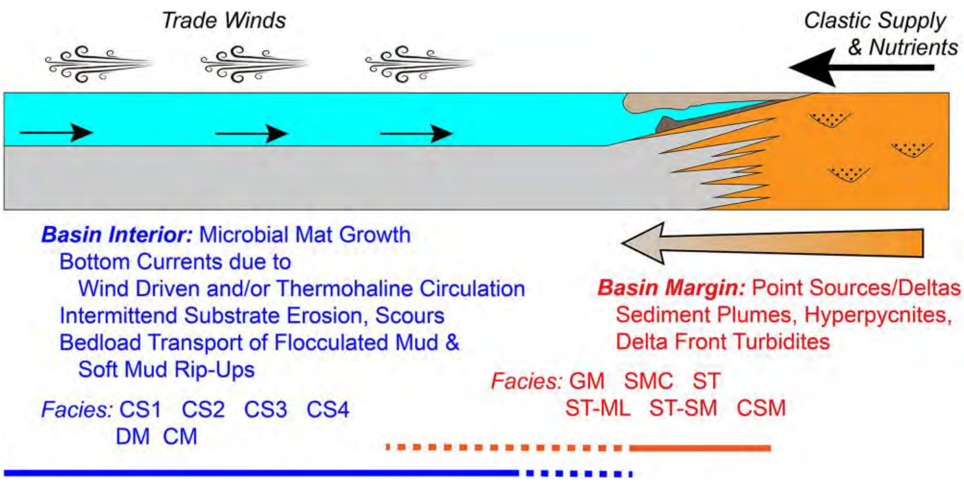

Figure 33. Conceptual view of sediment inputs, basin circulation, and distribution of facies types within the Amungee Member. CM = cemented mudstone; CS1 = striped carbonaceous shale; CS2 = faintly striped carbonaceous shale; CS3 = striped calcareous carbonaceous shale; CS4 = faintly striped calcareous carbonaceous shale; CSM = convolute silt/mud; DM = dark mudstone; GM = gray mudstone; SMC = silt-mud couplet; ST = siltstone; ST-ML = silt with muddy laminae; ST-SM = silt and soupy mud.

with laterally uniform black shale intervals (Figures 30, 31). This bipartite view of basin sedimentary dynamics is summarized in Figure 33. Sediment pulses introduced at the basin margin do not leave a distinct clastic signal (e.g., cyclic changes of grain size or detrital mineralogy) in the muds of the basin interior, and the best markers of sedimentation cycles are the slowly deposited DM and CM beds that likely represent parasequence boundaries and maximum flooding surfaces. In places where enough core intercepts are available, they should allow detailed long-distance correlations between basin margin and basin interior.

CONCLUSIONS

Initially drab gray shales and mudstones can be described in a systematic way on the basis of layering styles (core examination) and further details of sedimentary features and textures (thin section studies). The 12 lithologic subdivisions of the Amungee Member enable detailed and efficient core descriptions for lateral comparisons of depositional setting.

Amungee Member shales and mudstones contain a rich inventory of sedimentary features and fabrics. Through prior case studies and flume analogue investigations, physical, biological, and chemical processes that controlled transport, deposition, and lithification can be identified, allowing a detailed and realistic interpretation of depositional conditions for a succession deposited more than a billion years ago. The Amungee seafloor was covered with benthic microbial mats, and intermittently eroded by bottom currents that generated new sedimentary particles (mud floccules, mat fragments, water-rich mud rip-ups). The latter were transported across the basin floor and eventually redeposited as distinct facies (silt laminated shales, lenticular fabric). Along the basin margins, these "interior shales" interfinger and interact (via reworking) with mudstones that represent the distal ends of clastic wedges, supplied by gravity flow deposits (low-density turbidites, hyperpycnites) and riverine sediment plumes.

Microbial mats were the major source of kerogen for the thick and laterally continuous organic-rich shale intervals that made the Amungee Member the prime source rock for a world-class shale gas resource.

Basin fill and sedimentary architecture reflect the interplay of gravity-driven sediment supply from the basin margins with basin-wide wind- and/or thermohaline-driven bottom currents that spread fine-grained sediment over large distances across the basin interior.

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